WoundCareIQ — Product Requirements Document

Oct 18, 2026 · @Shokaib Haider

1. Executive summary

WoundCareIQ turns the wound assessments community teams already record into one prioritised worklist per caseload: every patient gets one priority (SEE TODAY, SEE THIS WEEK, MONITOR or HEALING), one recommended action, and a plain-English explanation a Tissue Viability Nurse (TVN) would recognise as their own reasoning.

WoundCareIQ does not replace SystmOne, EMIS or Net Health. It reads from them, consolidates every assessment per patient, analyses healing trajectory over 2 to 12 weeks, applies transparent clinical rules (DFU with PAD, infection, pressure ulcer category, chronicity, rapid growth), and attaches a Clinical Confidence rating that tells the clinician how much the data can be trusted.

Prioritisation is hybrid. Hard-coded clinical rules are the safety layer and set the minimum priority; an AI Clinical Reasoning Engine then reviews the patient's entire wound journey and writes a clinical narrative a TVN would recognise: why this patient, why now, what changed and what should happen next. The AI can escalate with cited evidence but can never lower a rule-based priority (Section 11.6).

Phase 1 is a CSV-based pilot with one NHS community provider, running in shadow mode alongside existing practice, with every recommendation reviewed by a TVN who marks it Agree or Disagree. That feedback is the validation dataset, the clinical safety evidence and the trust mechanism.

1.1 Document control

Item	Detail
Product	WoundCareIQ clinical intelligence platform
Document type	Product Requirements Document (single source of truth)
Version	1.0 (draft for pilot-site and engineering review)

Item	Detail
Status owner	Head of Product, WoundCareIQ
Clinical owner	Clinical Safety Officer (registered TVN)
Technical owner	CTO
Review cycle	Monthly during pilot; on every change to prioritisation rules
Change control	Rule changes require Clinical Safety Officer sign-off and a hazard log update (DCB0129)

1.2 Who should read what

Audience	Read first	Why
Software engineers, CTOs	Sections 5, 6, 7 to 11, 13, 14, 15	Data model, engines, APIs, schema, requirements
TVNs, community and district nurses	Sections 3, 4, 7 to 12, 16	Personas, clinical logic, explainability, workflow, validation
NHS community providers, pilot sites	Sections 12, 17, 18	Workflow change, compliance, pilot design
ICBs, Health Innovation Networks	Sections 2, 18, 19, 20	Problem, outcomes, pilot metrics, roadmap
Investors	Sections 1, 2, 20	Opportunity, differentiation, scale
Academic collaborators	Sections 8 to 10, 16, 18	Algorithms, confidence, validation methodology

1.3 Scope of this document

- **In scope:** clinical prioritisation of existing wound caseloads, trajectory analysis, explainability, TVN validation, CSV and EPR ingestion, reporting, compliance, pilot and roadmap.
- **Out of scope for v1:** image-based wound measurement, autonomous treatment selection, dressing formulary management, prescribing, direct patient-facing apps.

- **Clinical position:** WoundCareIQ is clinical decision support. It never makes a decision; a registered clinician always decides. This position, and its regulatory consequences, are set out in Section 17.

2. Mission, problem and vision

Mission: transform wound assessment data into actionable clinical intelligence, so community teams see the right patient, at the right time, with the right intervention.

2.1 The clinical and operational problem

The NHS managed an estimated 3.8 million patients with a wound in 2017/18 at an annual cost of £8.3 billion, and 81% of that cost fell in the community (Guest et al, BMJ Open 2020). Only 49% of chronic wounds healed within the study year, and unhealed wounds accounted for £5.6 billion of the cost. Wound care consumed 54.4 million district and community nurse visits that year.

The cost of an unhealed wound is several times that of a healed one. For a venous leg ulcer the mean annual cost per patient was about £7,900 unhealed versus £2,000 healed (Table 10). Every week a deteriorating wound goes unnoticed is expensive for the patient and the system.

Current systems record this activity well but do not interpret it. SystmOne, EMIS and Net Health hold assessments, measurements and care plans, yet none of them answer the four questions a community team lead asks every morning:

Question	What teams do today	What it costs
Who needs attention today?	Scan the caseload, rely on handover and memory	Urgent patients surface late or by chance
Who is deteriorating?	Compare measurements by hand across past entries	Gradual deterioration is missed between visits
Who should be escalated to TVN, vascular or the diabetic foot MDT?	Referral depends on individual nurse confidence	Variation in referral, delayed specialist input
Who can safely wait?	Visit frequency is set once and rarely revisited	Specialist and nurse time spent on healing wounds

2.2 Why this problem persists

1. **Data is longitudinal, screens are not.** EPRs show one assessment at a time. Trajectory, the strongest predictor of outcome, has to be reconstructed by the clinician.
2. **Caseloads exceed specialist capacity.** A single TVN may support hundreds of open wounds across a footprint and cannot review each one.
3. **Risk is combinatorial.** A diabetic foot ulcer is serious; a diabetic foot ulcer with peripheral arterial disease and new infection is limb-threatening. Systems store each fact separately.
4. **Reports count activity, not risk.** Existing reporting answers 'how many wounds' rather than 'which wounds are getting worse'.
5. **Data quality is uneven.** Missing measurements and free-text entries make simple dashboards unreliable, so teams stop trusting them.

2.3 What WoundCareIQ does

WoundCareIQ sits on top of existing systems as a read-only intelligence layer. It ingests assessments, consolidates them per patient, and returns one prioritised worklist with the reasoning visible. Clinicians stay in their EPR for documentation; WoundCareIQ tells them where to look first.

2.4 Vision

Build the clinical intelligence layer for wound care.

1. **Data collection** (existing EPRs): assessments recorded at the point of care.
2. **Clinical intelligence** (WoundCareIQ, Phases 1 to 3): consolidation, trajectory, prioritisation, explainability.
3. **Proactive intervention** (Phase 4): predicted deterioration flagged before it is visible in measurements.
4. **Better outcomes** (Phase 5): population analytics for ICBs; healing rates, time to specialist review and avoidable escalations tracked across a system.

Existing systems collect wound data. WoundCareIQ interprets it.

3. User personas

Eight personas use WoundCareIQ; the TVN is the primary design persona because the product must think like an experienced TVN, and every other persona benefits from that reasoning being made visible.

Persona	Primary job in WoundCareIQ	Key view	Access level
Tissue Viability Nurse	Triage specialist caseload, validate priorities	Specialist worklist, patient detail, feedback	Clinical, footprint-wide
Community Nurse	Plan daily visits	Team worklist, patient detail	Clinical, own team
District Nurse (team lead)	Allocate visits, escalate	Team worklist, escalation queue	Clinical, own team(s)
Care Home Nurse	Know which residents need escalation	Facility worklist	Clinical, own facility
Director of Nursing	Assure quality and safety	Organisation dashboard	Aggregate, drill-down with audit
Community Provider Manager	Match capacity to demand	Operational dashboard	Aggregate, pseudonymised
ICB	Commission and compare outcomes	Population analytics	Aggregate, anonymised
Wound Specialist / Surgeon	Receive the right referrals early	Referral queue, patient summary	Clinical, referred patients

3.1 Tissue Viability Nurse (TVN)

- **Context:** Band 7 or 8a specialist covering several community teams and care homes. Receives referrals by phone, email and task; often reviews notes before deciding whether to visit.
- **Needs:** a single view of every wound across the footprint ranked by clinical severity; trajectory without opening ten assessments; confidence that urgent patients are not buried.

- **Current workflow:** reads referral, opens EPR, scrolls through assessments, compares measurements by hand, phones the referring nurse, decides.
- **Challenges:** referral quality varies; no visibility of patients who should have been referred and were not; time lost on low-risk referrals.
- **Success criteria:** reviews SEE TODAY patients in under 10 minutes each morning; agrees with at least 85% of priorities; finds unreferred high-risk patients each week.
- **User story:** As a TVN, I want every patient on my footprint ranked by clinical severity with the reason shown, so that I review the most dangerous wounds first.

3.2 Community Nurse

- **Context:** Band 5 or 6, 6 to 10 visits a day, mixed caseload of which wounds are a large share.
- **Needs:** to know before setting out which patients are getting worse; prompts for missing assessments (for example no ABPI recorded on a leg ulcer).
- **Challenges:** limited time to analyse trends; continuity breaks when patients are seen by different nurses.
- **Success criteria:** sees 'what changed since last review' for every patient in one line; fewer surprises at the bedside.
- **User story:** As a community nurse, I want to see which of today's patients have deteriorated since the last visit, so that I can bring the right equipment and plan escalation.

3.3 District Nurse (team lead)

- **Context:** Band 6 or 7 caseload holder, allocates daily visits across the team, accountable for the caseload.
- **Needs:** a team worklist to allocate from; an escalation queue; visibility of overdue reassessments.
- **Challenges:** capacity pressure; deciding which visits can be safely deferred.
- **Success criteria:** daily allocation meeting uses WoundCareIQ as its agenda; zero SEE TODAY patients left unallocated.
- **User story:** As a district nurse, I want to know which patients can safely wait, so that I can protect capacity for those who cannot.

3.4 Care Home Nurse

- **Context:** registered nurse in a nursing home, often the only RN on shift; residents with pressure ulcers and leg ulcers; relies on community and TVN input.
- **Needs:** a clear escalation signal and a reason to give when calling for help.
- **Challenges:** variable access to EPRs; documentation may sit in a separate care-home system.
- **Success criteria:** escalations are made with a structured reason; fewer avoidable Category 3 and 4 pressure ulcers.
- **User story:** As a care home nurse, I want a clear flag when a resident's wound needs specialist review, so that I can escalate with confidence.

3.5 Director of Nursing (DoN)

- **Needs:** assurance that high-risk wounds are seen promptly; evidence for quality committees; early warning of harm (pressure ulcer incidence, delayed DFU referral).
- **Success criteria:** monthly board report generated from WoundCareIQ; time from SEE TODAY flag to clinical review reported and improving.
- **User story:** As a Director of Nursing, I want to see how quickly high-risk wounds are reviewed across all teams, so that I can assure the board on wound care safety.

3.6 Community Provider Manager

- **Needs:** demand by priority band per team; visit capacity versus need; trends in caseload complexity.
- **Success criteria:** capacity decisions use priority-weighted caseload rather than headcount of patients.
- **User story:** As a provider manager, I want caseload demand shown by clinical priority per team, so that I can move capacity to where risk is highest.

3.7 Integrated Care Board (ICB)

- **Needs:** population-level healing rates, chronicity profile, variation across providers, evidence that commissioned services improve outcomes; alignment with National Wound Care Strategy Programme (NWCSP) recommendations.
- **Success criteria:** quarterly anonymised outcome reports comparable across providers.

- **User story:** As an ICB commissioner, I want comparable healing and escalation metrics across providers, so that I can target investment and reduce unwarranted variation.

3.8 Wound Care Specialist / Surgeon (vascular, diabetic foot MDT, plastics)

- **Needs:** earlier, better-documented referrals; a summary showing trajectory, vascular status and infection history at a glance.
- **Success criteria:** referrals arrive with structured context; fewer late-presenting limb-threatening wounds.
- **User story:** As a vascular surgeon, I want referrals for DFU with PAD to arrive with trajectory and vascular data attached, so that I can triage them the same day.

4. Core product principles

Five principles govern every design decision; where a requirement conflicts with a principle, the principle wins and the requirement is revised.

#	Principle	What it means in the product	What it rules out
P1	Clinical severity determines priority	Priority comes from clinical rules: a DFU with PAD is SEE TODAY whether the caseload has 5 such patients or 50	Percentile cut-offs ('top 10%'), fixed quotas per band, normalising to caseload size
P2	One patient, one priority, one recommendation	All wounds and assessments roll up to a single patient row with the highest-acuity priority and one primary action	Duplicate patient rows, one row per assessment, conflicting recommendations
P3	Historical progression matters more than a single assessment	Trajectory across 2 to 12 weeks drives priority and confidence; the latest assessment is read in context	Ranking on the latest measurement alone

#	Principle	What it means in the product	What it rules out
P4	Every recommendation is clinically explainable	Each priority shows its primary driver, supporting factors and what changed, in clinical language	Opaque scores, unexplained machine-learning outputs in clinical views
P5	The system thinks like an experienced TVN	Rules are authored and signed off by TVNs; red flags override arithmetic; missing critical data is itself a finding	Purely statistical weighting with no clinical override

4.1 How the principles shape the algorithm

1. **Red flags first, scores second.** Hard clinical rules (for example DFU with PAD, spreading infection, Category 4 pressure ulcer) assign SEE TODAY directly. Weighted scoring only ranks patients within a band and decides bands where no red flag fires.
2. **Absolute thresholds, never relative.** Band boundaries are fixed clinical score thresholds. If every patient on a caseload is high risk, every patient is shown as high risk.
3. **Highest wound wins.** A patient's priority is the most severe priority across their open wounds; other wounds appear as supporting factors.
4. **Uncertainty is shown, not hidden.** Low Clinical Confidence never lowers priority. It adds a 'data needed' action (for example 'record ABPI').
5. **Fail safe.** If the engine cannot compute a priority, the patient is shown as 'Unable to prioritise: clinical review required', never silently dropped.
6. **Hybrid reasoning, rules as the floor.** Deterministic rules decide the minimum priority. The AI Clinical Reasoning Engine adds whole-journey judgement on top and may escalate, never de-escalate. If the AI is unavailable or fails validation, the rules and their explanation stand alone (Section 11.6).

5. Supported data sources and integration

WoundCareIQ is source-agnostic: every source is mapped into one canonical wound data model (Section 6), so the engines never see a supplier format. Phase 1 uses CSV exports; Phases 2 and 3 add direct SystmOne and EMIS feeds; later phases add FHIR-based EPRs.

Source	Typical setting	Phase 1 route	Target route	Notes
SystmOne (TPP)	Community nursing, some care homes	Scheduled report export to CSV	Supplier-approved interface via TPP partner route; nightly then near-real-time	Wound templates are often locally configured per trust, so mapping is per site
EMIS Web	General practice, practice nurses	Search and Reports export to CSV	Supplier-approved interface via EMIS partner route	Wound data often coded with SNOMED CT plus free text
Net Health Wound Care	Specialist wound clinics	CSV or API export	Vendor REST API	Richer structured measurements and images; images out of scope in v1
Future EPRs (acute and community)	Trusts, care-home platforms	CSV	HL7 FHIR R4 UK Core	Observation, Condition, Encounter resources

Access routes and their commercial terms must be confirmed with each supplier before Phase 2 and Phase 3 commitments; this is tracked as open question OQ-1 in Section 21.

5.1 Required fields

Fields are graded by importance. **Critical** fields are needed to prioritise at all; **important** fields change priority; **enriching** fields raise confidence.

Field	Grade	Used by	Example
NHS number	Critical	Consolidation	943 476 5919 (validated with modulus 11)
Assessment date and time	Critical	Trajectory, chronicity	2026-10-14 10:32
Wound identifier or location	Critical	Wound grouping	Left heel; wound 2
Wound type / aetiology	Critical	Rules, weighting	Diabetic foot ulcer; venous leg ulcer; pressure ulcer
Length, width (cm)	Important	Area, trajectory	3.2 × 2.1
Depth (cm)	Important	Trajectory, severity	0.4
Pressure ulcer category	Important	Rules	Category 3
Infection signs / status	Important	Infection logic	Increasing erythema, malodour, pain
Wound onset date	Important	Chronicity	2026-06-02
Diabetes status	Important	DFU logic	Type 2
PAD indicators	Important	DFU+PAD logic	ABPI 0.62; absent foot pulses; toe pressure
Tissue type percentages	Enriching	Trajectory	Slough 40%, granulation 60%
Exudate level and type	Enriching	Infection, trajectory	Heavy, purulent
Pain score	Enriching	Infection	7/10, increasing
Wound bed, edge, periwound	Enriching	Trajectory	Undermining 1 cm at 12 o'clock
Mobility, nutrition (MUST), continence	Enriching	Risk factors	MUST 2
Treatment, dressing, compression	Enriching	Recommendations	Compression 40 mmHg

Field	Grade	Used by	Example
Assessing clinician role, team, facility	Important	Routing, audit	Band 6, Team North

5.2 Mapping strategy

1. **Site mapping profile.** Each source instance has a versioned mapping profile (YAML) that maps local template fields, codes and picklist values to canonical fields.
2. **Terminology.** SNOMED CT is the canonical terminology for wound type, location, infection signs and conditions. Local Read v2 or CTV3 codes and picklists are mapped via lookup tables maintained by WoundCareIQ clinical informaticians.
3. **Units and normalisation.** Measurements converted to centimetres; area computed as length × width with a configurable elliptical correction (× 0.785) recorded on the record so trajectories never mix methods.
4. **Free text.** In v1 free text is stored but not used for prioritisation, except a curated keyword list for infection and red flags, shown to the user as 'from free text'.
5. **Mapping validation.** A new site goes live only after 200 sampled assessments are dual-checked by a WoundCareIQ informatician and a site TVN, with at least 98% field-level agreement on critical fields.

5.3 Data quality challenges and handling

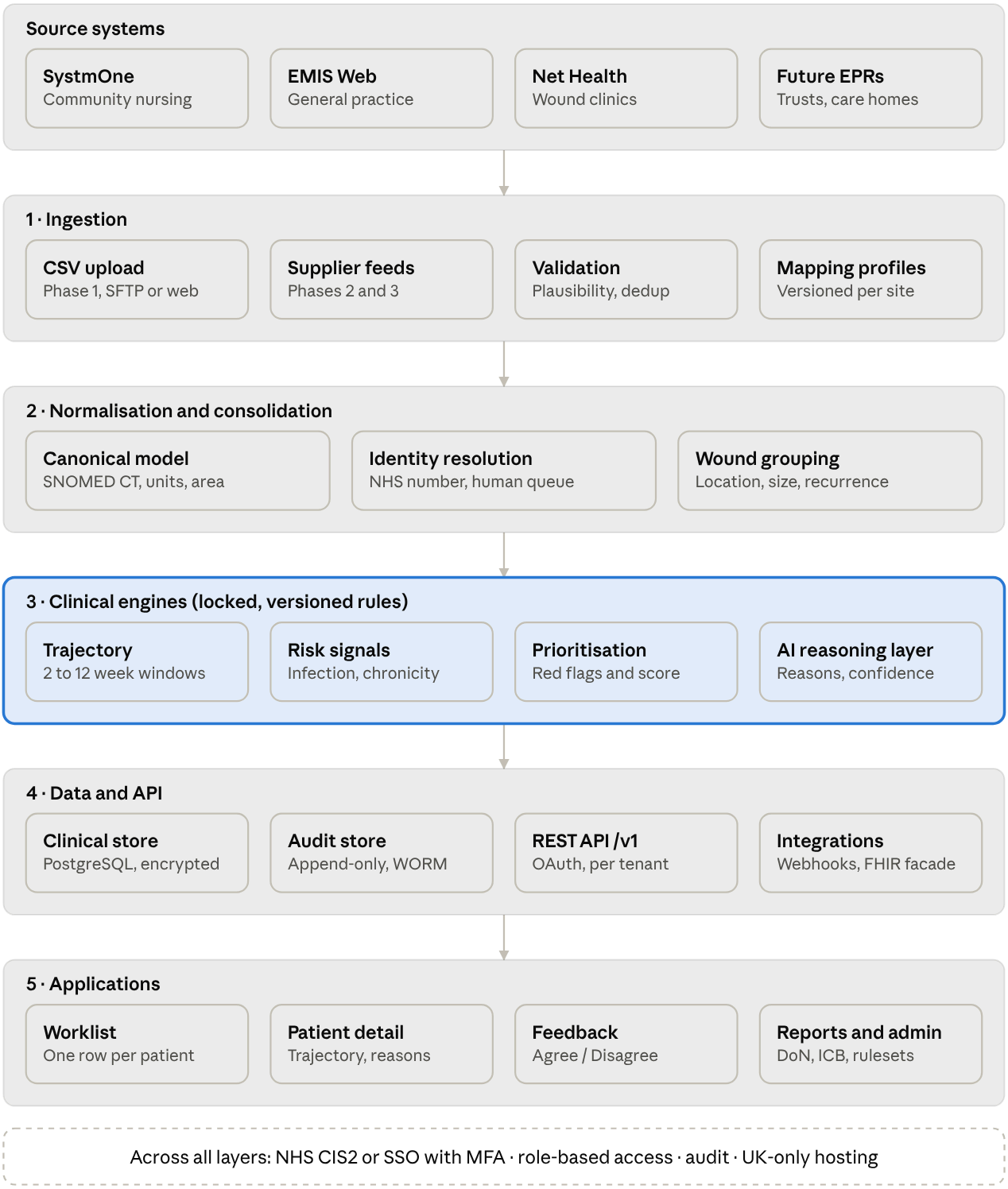
Challenge	Example	Handling
Missing measurements	Assessment recorded with no size	Trajectory uses nearest valid points; confidence lowered; 'measure wound' action added
Inconsistent wound identity	'L leg', 'left gaiter', 'LLL' for the same wound	Location normalisation plus wound-matching rules (Section 7)
Duplicate assessments	Same assessment exported twice	Hash of source ID, patient, wound, timestamp and values; exact duplicates dropped
Late or back-dated entries	Assessment entered 3 days after visit	Clinical date used for trajectory; entry date kept for audit

Challenge	Example	Handling
Implausible values	Length 32 cm instead of 3.2 cm	Plausibility rules (more than 300% change in a week, or outside type-specific bounds) flag the value; it is excluded from trajectory and shown to the user
Template drift	Trust changes its wound template	Mapping profile versioned; unmapped fields raise a data-ops alert
No NHS number	Care-home record	Probabilistic match on name, date of birth, postcode, held for human confirmation; never auto-merged

5.4 Integration architecture

The platform runs in five layers, each separately deployable and testable: ingestion, normalisation, the clinical engines, an API layer and the user-facing applications. All processing runs in UK-hosted cloud infrastructure (UK South / UK West regions).

Every source is mapped to one model before the engines see it



WoundCareIQ integration architecture · 5 layers

Data flows one way, from source systems down to the applications; WoundCareIQ never writes back to the EPR in v1, so the EPR stays the record of care. Engines run on the canonical model only, which is why adding a new source changes mapping profiles, not clinical logic.

6. Data model

The canonical model has eight core entities: a Patient has many Wounds, each Wound has many Assessments, and the engines write Risk Factors, Recommendations, Prioritisation Events and Clinical Confidence records against the Patient. Facility describes where care happens. All engine outputs are immutable and versioned so any past worklist can be reproduced exactly.

Entity	Purpose	Cardinality	Written by
Patient	One consolidated person	1 per NHS number per tenant	Consolidation engine
Facility	Team, care home, clinic	Many patients	Admin / ingestion
Wound	One anatomical wound over its lifetime	Many per patient	Consolidation engine
Assessment	One clinical assessment of one wound	Many per wound	Ingestion
Risk Factor	Patient-level condition or status (diabetes, PAD, MUST)	Many per patient, time-bounded	Ingestion and engines
Recommendation	The one action for the patient now	1 current per patient	Prioritisation engine
Prioritisation Event	A computed priority with full reasoning, immutable	Many per patient (history)	Prioritisation engine
Clinical Confidence	Confidence rating and its components	1 per prioritisation event	Confidence engine

6.1 Patient

```
{
  "patient_id": "pat_01HF8Z3K9Q",
  "tenant_id": "org_humber_community",
  "nhs_number": "9434765919",
  "nhs_number_verified": true,
  "date_of_birth": "1946-03-11",
  "sex": "F",
  "facility_id": "fac_team_north",
  "care_setting": "home",
  "source_patient_refs": [
    {"source": "systmone", "ref": "S1-88123"},
    {"source": "emis", "ref": "E-55001"}
  ],
  "open_wound_count": 2,
  "current_priority": "SEE_TODAY",
  "current_prioritisation_event_id": "pe_01HG2A7M",
  "created_at": "2026-09-01T08:00:00Z",
  "updated_at": "2026-10-15T06:02:11Z"
}
```

6.2 Wound

```
{
  "wound_id": "wnd_01HF9B",
  "patient_id": "pat_01HF8Z3K9Q",
  "wound_type": "diabetic_foot_ulcer",
  "snomed_code": "280137006",
  "location": {"side": "left", "site": "plantar_first_metatarsal_head"},
  "onset_date": "2026-07-20",
  "onset_date_source": "recorded",
  "status": "open",
  "pressure_ulcer_category": null,
  "first_assessment_at": "2026-07-22T10:00:00Z",
  "latest_assessment_at": "2026-10-14T10:32:00Z",
  "assessment_count": 9
}
```

6.3 Assessment

```
{
  "assessment_id": "asm_01HG1X",
  "wound_id": "wnd_01HF9B",
  "source": {"system": "systmone", "record_id": "S1-A-449201",
"mapping_profile": "humber-s1-v3"},
  "assessed_at": "2026-10-14T10:32:00Z",
  "entered_at": "2026-10-14T16:05:00Z",
  "assessor_role": "band_6_community_nurse",
  "measurements": {"length_cm": 2.8, "width_cm": 2.2, "depth_cm": 0.5,
"area_cm2": 6.16, "area_method": "lxw"},
  "tissue": {"granulation_pct": 30, "slough_pct": 60, "necrotic_pct": 10},
  "exudate": {"level": "moderate", "type": "seropurulent"},
  "infection_signs": ["increased_pain", "erythema_lt_2cm", "malodour"],
  "infection_status_recorded": "suspected",
  "pain_score": 6,
  "vascular": {"abpi": 0.58, "foot_pulses": "absent", "toe_pressure_mmhg":
null},
  "quality_flags": [],
  "free_text": "Wound edge macerated..."
}
```

6.4 Facility

```
{
  "facility_id": "fac_team_north",
  "tenant_id": "org_humber_community",
  "type": "community_nursing_team",
  "name": "Team North",
  "ods_code": "RV9XX",
  "parent_facility_id": "fac_locality_east",
  "tvn_cover_ids": ["usr_tvn_04"]
}
```

6.5 Risk Factor

```
{
  "risk_factor_id": "rf_01HF",
  "patient_id": "pat_01HF8Z3K9Q",
  "type": "peripheral_arterial_disease",
  "evidence": {"kind": "abpi", "value": 0.58, "assessment_id": "asm_01HG1X"},
  "severity": "moderate",
  "valid_from": "2026-10-14",
  "valid_to": null,
  "derived": true
}
```

6.6 Prioritisation Event

```
{
  "prioritisation_event_id": "pe_01HG2A7M",
  "patient_id": "pat_01HF8Z3K9Q",
  "computed_at": "2026-10-15T06:02:11Z",
  "engine_version": "prio-1.4.0",
  "ruleset_version": "rules-2026.10.1",
  "input_snapshot_hash": "sha256:9f1c...",
  "priority": "SEE_TODAY",
  "previous_priority": "SEE_THIS_WEEK",
  "clinical_score": 87,
  "red_flags_fired": ["RF-01_DFU_WITH_PAD", "RF-
05_INFECTION_WITH_DETERIORATION"],
  "primary_driver": "DFU with PAD (ABPI 0.58) and suspected infection",
  "supporting_factors": ["Area +38% over 2 weeks", "Slough increased 20% to
60%", "Wound open 87 days"],
  "changes_since_last_review": ["New ABPI 0.58 recorded", "Infection signs
newly present"],
  "trajectory": {"wound_id": "wnd_01HF9B", "class": "DETERIORATING"},
  "recommendation_id": "rec_01HG2A",
  "confidence_id": "cc_01HG2A"
}
```

6.7 Recommendation and Clinical Confidence

```
{
  "recommendation_id": "rec_01HG2A",
  "action_code": "URGENT_REFERRAL_DIABETIC_FOOT_MDT",
  "action_text": "Same-day referral to diabetic foot MDT / vascular; review
for infection today",
  "secondary_actions": ["Record toe pressure if available"],
  "due_by": "2026-10-15T17:00:00+01:00"
}
```

```
{
  "confidence_id": "cc_01HG2A",
  "level": "HIGH",
  "score": 82,
  "components": {"assessment_count": 25, "completeness": 22,
"trajectory_reliability": 20, "recency": 15},
  "limiting_factors": ["No toe pressure recorded"]
}
```

6.8 AI Review

One AI Review is stored per prioritisation event, linked by `prioritisation_event_id`, whether it was displayed or rejected.

```
{
  "ai_review_id": "air_01HG2A",
  "prioritisation_event_id": "pe_01HG2A7M",
  "bundle_hash": "sha256:4be0...",
  "model_version": "reasoner-1.0.0",
  "prompt_version": "tvn-review-2026.10",
  "rule_priority": "SEE_TODAY",
  "ai_escalation": null,
  "final_priority": "SEE_TODAY",
  "validation": {"status": "passed", "checks_failed": [], "attempts": 1},
  "outputs": {"current_status": "...", "primary_driver": "...",
"what_changed": ["..."], "recommended_action": "...", "clinical_narrative":
"..."},
  "displayed": true,
  "generated_at": "2026-10-15T06:02:48Z"
}
```

7. Patient consolidation engine

The consolidation engine guarantees one row per patient: every assessment from every source is used, grouped by patient and then by wound, and produces exactly one patient record, one prioritisation decision and one recommended action. Historical assessments feed trajectory, risk and confidence; they never appear as duplicate rows.

7.1 Pipeline

1. **Identity resolution.** Group records by validated NHS number within a tenant. Records without one go to a probabilistic match (surname, date of birth, sex, postcode) and are held in a 'confirm identity' queue for a human; they are never auto-merged.
2. **Deduplication.** Drop exact duplicates by content hash. Near-duplicates (same wound, same clinician, within 30 minutes, values differing only by rounding) are merged, keeping the more complete record.
3. **Wound grouping.** Assign each assessment to a wound using, in order: source wound identifier; normalised anatomical location plus side plus wound type; then size continuity (a new assessment within 50% area of the open wound at that site). Ambiguous matches create a 'possible new wound' flag rather than silently merging.
4. **Wound lifecycle.** A wound recorded healed, then re-recorded at the same site within 90 days, is linked as a **recurrence** of the same wound (relevant to chronicity and the 'repeated deterioration' class). After 90 days it is a new wound with a recurrence risk factor.
5. **Per-wound analysis.** Trajectory (Section 8), infection status and chronicity are computed per wound.
6. **Patient roll-up.** The patient's priority is the highest priority across open wounds. The primary driver comes from the driving wound; other wounds contribute supporting factors. One recommendation is selected (Section 9.7).
7. **Event write.** A new Prioritisation Event is written only when inputs change; unchanged inputs keep the existing event, so history shows real changes, not nightly noise.

7.2 Requirements

ID	Requirement	Priority
CON-01	The worklist shall contain at most one row per patient per tenant	Must

ID	Requirement	Priority
CON-02	Every ingested assessment shall be linked to exactly one wound or to a review queue	Must
CON-03	Patients shall never be auto-merged without a validated NHS number match	Must
CON-04	Patient priority shall equal the most severe priority across open wounds	Must
CON-05	A patient with multiple open wounds shall show the wound count and each wound's trajectory on the detail view	Must
CON-06	Users with the TVN role shall be able to split or merge wounds, with the change audited and engines re-run	Should
CON-07	Consolidation of a 5,000-patient caseload shall complete in under 5 minutes	Should

7.3 Acceptance criteria

- **Given** a patient with 12 assessments across 2 wounds from SystmOne and EMIS, **when** the import completes, **then** the worklist shows one row, the patient detail shows 2 wounds, and all 12 assessments appear in wound history.
- **Given** two records with the same name and date of birth but no NHS number, **when** imported, **then** no merge occurs and an identity-confirmation task is created.
- **Given** a patient with one healing venous leg ulcer and one deteriorating heel pressure ulcer, **when** prioritised, **then** the row shows the pressure ulcer's priority and driver, with the leg ulcer listed as a supporting wound.

8. Healing trajectory engine

The trajectory engine classifies every open wound as Improving, Stable, Delayed Healing, Deteriorating or Repeated Deterioration by comparing the latest assessment with the wound's state 2, 4, 6, 8 and 12 weeks earlier, using area as the primary signal and tissue, depth, exudate, pain and infection as modifiers. Short windows catch acute deterioration; long windows catch wounds that are quietly failing to heal.

8.1 Analysis windows

Window	Baseline tolerance	Clinical purpose
2 weeks	±3 days	Acute change; aligns with the NWCSP expectation that a lower-limb wound not healed in 2 weeks gets a full assessment
4 weeks	±5 days	The key prognostic point: area reduction at 4 weeks predicts healing by 12 weeks
6 weeks	±7 days	Confirms or refutes the 4-week signal
8 weeks	±7 days	Identifies plateau after early progress
12 weeks	±10 days	Hard-to-heal threshold; drives specialist review

A window is **valid** if an assessment with a usable area exists within tolerance of its baseline date. If not, the engine uses the nearest valid assessment within twice the tolerance and marks the window 'interpolated', lowering confidence.

8.2 Core measures

Percentage area reduction for window w:

```
\mathrm{PAR}_w = \frac{A_{w} - A_{now}}{A_{w}} \times 100
```

Positive PAR means the wound is shrinking. Where three or more valid measurements exist in a window, the engine also fits a robust (Theil–Sen) slope to log area over time, which resists a single mis-measured value:

```
k = \operatorname{median}_{i<j} \frac{\ln A_j - \ln A_i}{t_j - t_i}
```

k below zero is healing; above zero is growth. Weekly rate = (e^(7k) − 1) × 100%.

8.3 Type-specific healing benchmarks

The 4-week benchmark follows widely used evidence that wounds failing to reach a defined area reduction by week 4 are unlikely to heal by week 12. Values are configurable per site and signed off by the Clinical Safety Officer.

Wound type	Expected PAR at 4 weeks	Expected PAR at 12 weeks
Diabetic foot ulcer	⩾ 50%	Healed or ⩾ 90%
Venous leg ulcer	⩾ 40%	Healed or ⩾ 80%
Pressure ulcer Category 2	⩾ 40%	Healed
Pressure ulcer Category 3 or 4	⩾ 30%	⩾ 60%
Surgical wound (secondary intention)	⩾ 40%	Healed
Other / unspecified	⩾ 40%	⩾ 75%

8.4 Classification rules

Rules are evaluated in this order; the first match wins.

Order	Class	Rule (any condition triggers unless stated)
1	**Repeated Deterioration**	Two or more separate deterioration episodes in the last 12 weeks (an episode is a Deteriorating classification followed by at least one non-deteriorating assessment), or a recurrence at the same site within 90 days of recorded healing
2	**Deteriorating**	Area up ⩾ 10% over 2 weeks or ⩾ 20% over 4 weeks; depth up ⩾ 0.3 cm; necrotic or slough tissue up ⩾ 20 percentage points; pressure ulcer category increased; infection status escalated (Section 9.4); new undermining or tracking
3	**Delayed Healing**	Wound open ⩾ 4 weeks and 4-week PAR below the type benchmark, with no deterioration trigger; or 12-week PAR below the 12-week benchmark
4	**Stable**	Area change between −benchmark and +10% and wound open < 4 weeks (too early to judge); or care goal recorded as maintenance or palliative with no deterioration trigger

Order	Class	Rule (any condition triggers unless stated)
5	**Improving**	4-week PAR at or above benchmark, or 2-week PAR ⩾ 20%, with no worsening of tissue, exudate or infection
—	**Insufficient data**	Fewer than 2 assessments with area, and no qualitative deterioration trigger; the patient is prioritised on other factors and confidence is LOW

8.5 Modifiers from non-area signals

Area can mislead: debridement makes a wound larger while improving it, and a wound can shrink while becoming infected. The engine applies these modifiers:

- **Post-debridement grace.** If sharp or mechanical debridement is recorded, area increase in the following 14 days does not trigger Deteriorating unless tissue or infection also worsens.
- **Qualitative override.** Increasing pain, new malodour, spreading erythema or new necrosis classify the wound as Deteriorating even when area is static or falling.
- **Measurement noise.** Changes under 0.5 cm² on wounds under 2 cm² are treated as no change.
- **Implausible values.** Flagged values (Section 5.3) are excluded and listed in the explanation.

8.6 Worked example

Date	Area (cm²)	Slough	Note
2026-07-22	8.0	20%	First assessment, venous leg ulcer, compression started
2026-08-19	6.1	20%	4-week PAR 24% (below 40% benchmark)
2026-09-16	5.6	30%	8-week PAR 30%
2026-09-30	5.9	35%	Plateau
2026-10-14	6.6	40%	2-week area +12%

The 2-week area increase of 12% meets the Deteriorating rule, which takes precedence over Delayed Healing. The explanation reads: 'Deteriorating: area up 12% in 2 weeks after 12 weeks of slow progress (12-week reduction 18% against an expected 80%); slough increasing.'

9. Prioritisation engine

The prioritisation engine assigns each patient one of four priorities in two stages: clinical red flags place a patient in SEE TODAY directly, and a 0 to 100 clinical score with fixed thresholds places everyone else. Band membership never depends on how many other patients are in the band.

Priority	Meaning	Default response	Typical drivers
SEE TODAY	Risk of limb loss, sepsis or serious harm if not reviewed today	Clinical review or specialist referral the same working day	DFU with PAD, spreading infection, infection with deterioration, Category 4 pressure ulcer, rapid growth
SEE THIS WEEK	Not healing as expected or moderately worse	Review, plan change or TVN input within 5 working days	Delayed healing, chronic wound without recent specialist review, moderate deterioration, suspected infection
MONITOR	Stable, plan appropriate	Continue planned visits; reassess at next scheduled visit	Stable wound, maintenance goal
HEALING	Improving at or above expected rate	Continue plan; consider reducing visit frequency or step-down	Improving trajectory, no infection, no red flag

9.1 Decision sequence

1. **Safety screen.** If inputs suggest an emergency outside community scope (suspected acute limb ischaemia, signs of sepsis recorded), show a banner directing the clinician to emergency pathways (999 or same-day emergency referral, local sepsis protocol).

The product does not manage emergencies; it makes sure they are not missed in the list.

2. **Red flags.** Evaluate red-flag rules (9.2) on every open wound. Any fired red flag sets the wound to SEE TODAY.
3. **Clinical score.** Compute the 0 to 100 score for every wound (9.3).
4. **Band thresholds.** Map score to band, applying floors from trajectory and infection (9.4).
5. **Patient roll-up.** Patient priority = highest wound priority (Section 7).
6. **Acknowledgement check.** Apply any active TVN acknowledgement (9.4.2).
7. **Ordering within band.** Red-flag severity, then clinical score, then days since last clinical review, highest first.
8. **Recommendation.** Select the one primary action from the action library (9.8).
9. **AI clinical review.** The AI Clinical Reasoning Engine receives the wound journey bundle and the rule output, writes the six outputs and may propose a one-band escalation. The guardrail validator checks the result; final priority = the higher of rule priority and any validated escalation (Section 11.6).

9.2 Red-flag rules (SEE TODAY)

ID	Red flag	Trigger definition	Primary action
RF-01	DFU with PAD	Open diabetic foot ulcer and any PAD evidence (Section 9.5)	Same-day referral to diabetic foot MDT and vascular service
RF-02	Spreading or systemic infection	Erythema ≥ 2 cm beyond wound edge, lymphangitis, crepitus, or systemic signs recorded	Same-day medical review; follow local sepsis pathway if systemic
RF-03	Category 4, unstageable or deep tissue pressure ulcer	Newly recorded, deteriorating, or no TVN review in the last 7 days	TVN review today; check pressure-relieving equipment and repositioning plan
RF-04	Rapid wound growth	Area up ≥ 50% within 14 days or ≥ 25% within 7 days (outside post-debridement grace)	Clinical review today; reassess aetiology and infection

ID	Red flag	Trigger definition	Primary action
RF-05	Infection with deterioration	Suspected or active infection and trajectory Deteriorating	Clinical review today; consider wound swab and antimicrobial review
RF-06	DFU limb-threat signs	DFU with new necrosis or gangrene, exposed bone or positive probe-to-bone	Same-day diabetic foot MDT referral
RF-07	New DFU not under foot service	Active DFU with no recorded referral to the MDT foot service	Refer to MDT foot service within 1 working day, in line with NICE NG19
RF-08	Severe ischaemia, any lower-limb wound	ABPI < 0.5 or toe pressure < 30 mmHg, or rest pain recorded	Urgent same-day vascular referral

Red-flag thresholds are configurable per tenant only with Clinical Safety Officer approval, and every change is recorded in the hazard log.

9.3 Clinical score

The score is the sum of six components, each capped, so no single factor can dominate except through a red flag.

Component	Max	Scoring
Wound severity	25	PU Cat 4 / unstageable 25; DFU 20; PU Cat 3 18; arterial or mixed ulcer 16; VLU 10; PU Cat 2 8; surgical dehisced 12; other 6. Depth > 1 cm +3, undermining +2 (within cap)
Trajectory	25	Repeated Deterioration 25; Deteriorating 20; Delayed Healing 12; Insufficient data 8; Stable 5; Improving 0
Infection	20	Systemic or spreading 20; active local 14; suspected 9; resolving on treatment 6; resolved < 14 days 3; none 0

Component	Max	Scoring
Vascular and diabetes	15	DFU with PAD 15; PAD with leg ulcer 10; DFU without PAD 8; leg ulcer with no ABPI in 12 weeks 4; diabetes with non-foot wound 3
Chronicity	10	Section 9.7
Patient risk factors	5	+2 MUST ⩾ 2; +1 immobility (Braden ⩽ 12 or bed-bound); +1 immunosuppression; +1 renal failure on dialysis (cap 5)

```
S = \min(25, S_{sev}) + S_{traj} + S_{inf} + S_{vasc} + S_{chron} + \min(5, S_{risk})
```

9.4 Band thresholds and floors

Rule	Result
Any red flag fired	SEE TODAY
Score ⩾ 65	SEE TODAY
Score 40 to 64	SEE THIS WEEK
Score < 40 and trajectory Improving, no active or suspected infection	HEALING
Score < 40, otherwise	MONITOR
Floor: trajectory Deteriorating or Delayed Healing	Never below SEE THIS WEEK
Floor: suspected or active infection	Never below SEE THIS WEEK
Floor: pressure ulcer Category 4, unstageable or deep tissue injury	Never below SEE THIS WEEK
Floor: wound open > 90 days with no TVN review in 60 days	Never below SEE THIS WEEK
Engine error or no computable inputs	'Unable to prioritise: clinical review required', shown above MONITOR

9.4.1 Why absolute thresholds

Thresholds were set so that the bands reflect clinical meaning: a score of 65 cannot be reached without at least two serious components (for example Deteriorating trajectory 20, active infection 14, DFU 20, chronicity 6, risk factors 5). They are validated in the pilot against TVN judgement (Section 16) and adjusted only through clinical governance.

9.4.2 Red-flag acknowledgement

A patient already on an agreed daily plan should not dominate SEE TODAY indefinitely. A TVN may record 'Plan in place' against a red flag, which shows the patient as SEE THIS WEEK with an 'acknowledged' marker for up to 7 days (72 hours for RF-01, RF-02, RF-06 and RF-08). Any new deterioration, new red flag or expiry restores SEE TODAY automatically. Acknowledgements are audited and reported.

9.5 DFU + PAD logic

A diabetic foot ulcer with peripheral arterial disease is the highest-weighted combination in the engine because it is the commonest route to major amputation in community wound care, and it is the combination most easily under-recognised.

Why the combination is high risk

- **Reduced perfusion stops healing.** Oxygen and antibiotic delivery to the wound bed fall, so standard offloading and dressings fail.
- **Infection spreads faster and looks milder.** Ischaemia blunts the inflammatory response, so redness and swelling understate severity.
- **Neuropathy hides pain.** Patients often feel little, so neither they nor carers raise alarm.
- **Tests mislead.** Arterial calcification in diabetes can give a falsely normal or raised ABPI; pulses may be hard to assess in oedema.
- **Time matters.** Delays of days in revascularisation assessment change limb outcomes; NICE NG19 sets referral within 1 working day for any active diabetic foot problem and immediate referral for limb-threatening presentations.

PAD evidence used by the engine

Evidence	Interpretation	PAD tier
ABPI < 0.5, toe pressure < 30 mmHg, rest pain, gangrene	Severe ischaemia	Severe (also fires RF-08)

Evidence	Interpretation	PAD tier
ABPI 0.5 to 0.89	PAD present	Moderate
TBPI < 0.7	PAD present	Moderate
Absent or monophasic foot pulses on Doppler	PAD suspected	Suspected
ABPI > 1.3	Non-compressible vessels; PAD not excluded	Suspected (action: obtain toe pressure)
Coded diagnosis of PAD or previous revascularisation	PAD known	Moderate
ABPI 0.9 to 1.3 with triphasic pulses	PAD not evident	None

Weighting

- RF-01 fires at Suspected tier or above: the patient is SEE TODAY regardless of score.
- The vascular component scores the maximum 15, and wound severity scores DFU at 20, so a DFU with PAD scores at least 35 before trajectory or infection; with active infection and a deteriorating trajectory it reaches 69 on score alone.
- Severe tier additionally fires RF-08 and sorts above all other SEE TODAY patients except emergency banners.

Prioritisation behaviour

- Cannot be lowered to HEALING by an improving trajectory; the most it can fall to is SEE THIS WEEK through a 72-hour acknowledgement with MDT or vascular involvement recorded.
- Missing vascular assessment on any DFU is itself a finding: 'No vascular assessment recorded' is a supporting factor, confidence drops, and the recommendation includes 'Record foot pulses and toe pressure or ABPI'.

9.6 Infection logic

Infection is modelled as a state per wound, derived from recorded signs, the clinician's recorded status, antimicrobial prescribing and swab results. The state model follows the wound infection continuum used in international consensus guidance (contamination, colonisation, local, spreading, systemic) and NICE NG152 for leg ulcer infection.

State	Entry criteria	Score	Band effect
None	No signs, no recorded infection	0	None
Suspected	Subtle signs: increasing pain, delayed healing with friable granulation, malodour, increased exudate, pocketing; or clinician records 'query infection'	9	Floor SEE THIS WEEK
Active (local)	Overt local signs: erythema < 2 cm, warmth, swelling, purulent discharge; or clinician records infection; or antimicrobial started for the wound	14	Floor SEE THIS WEEK; RF-05 if deteriorating
Spreading	Erythema ⩾ 2 cm, lymphangitis, crepitus, wound breakdown with satellite lesions	20	RF-02, SEE TODAY
Systemic	Fever, rigors, confusion or other systemic signs recorded with the wound	20	RF-02 plus sepsis banner
Resolving	On antimicrobials, signs reducing on consecutive assessments	6	Floor SEE THIS WEEK until course ends
Resolved	No signs for ⩾ 14 days after treatment	3 for 14 days, then 0	None; counts towards infection history

Infection progression is the movement between states over time:

- **Escalation** (for example Suspected to Active, or Active to Spreading within any 14 days) is a deterioration trigger in the trajectory engine and is always listed under 'What changed'.
- **Failure to respond:** Active or Resolving for more than 14 days on antimicrobials without improvement adds 'Infection not responding to treatment: review antimicrobial and consider swab or specialist input'.

- **Recurrent infection:** two or more Active episodes in 12 weeks on the same wound adds 6 points to trajectory (within its cap) and contributes to the Repeated Deterioration class.

9.7 Chronicity engine

Chronicity measures how long the wound has been open, from recorded onset date or, if missing, the first assessment date (marked 'estimated'). Weight rises with duration because the chance of healing falls and the chance of a missed underlying cause rises.

Duration	Points	Clinical rationale	Additional behaviour
0 to 30 days	0	Acute phase; most wounds are expected to progress; trajectory is the better signal	Lower-limb wounds not healed by day 14 prompt 'complete full lower-limb assessment'
31 to 90 days	3	Past the 4-week prognostic point; failing wounds should already show as Delayed Healing	Prompt for aetiology confirmation if not recorded
91 to 180 days	6	Hard-to-heal; specialist review expected	Floor SEE THIS WEEK if no TVN review in 60 days
181 to 365 days	8	Chronic; cost and quality-of-life burden high; treatment plan likely needs to change	Prompt for care-goal review (healing versus maintenance)
365+ days	10	Long-standing non-healing; consider atypical aetiology, including malignancy	Prompt 'consider referral for biopsy or specialist opinion if not done'

Chronicity never sets SEE TODAY on its own. It ensures that long-standing wounds which have become 'part of the furniture' are surfaced for specialist review.

9.8 Recommendation selection

Each patient receives one primary recommendation from a controlled action library, chosen from the driving wound's highest-ranked trigger. Up to two secondary actions may be shown, typically data-capture actions.

Driver	Primary action	Typical secondary action
RF-01, RF-06, RF-07	Refer to diabetic foot MDT (same day for RF-01, RF-06; within 1 working day for RF-07)	Record vascular assessment
RF-02	Same-day medical review; sepsis pathway if systemic	Wound swab per local policy
RF-03	TVN review today	Check pressure-relieving equipment
RF-04, RF-05	Clinical review today	Reassess aetiology; review antimicrobials
RF-08	Urgent same-day vascular referral	Avoid compression until vascular review
Delayed Healing	TVN review this week; review treatment plan	Confirm aetiology and ABPI
Chronic, no recent TVN review	TVN case review this week	Care-goal review
Suspected infection	Review for infection this week	Increase assessment frequency
Stable	Continue plan	Measure at next visit
Improving	Continue plan; consider reduced visit frequency	None

Recommendations state what and when, never which dressing or drug: treatment choice remains clinical.

9.9 Worked prioritisation examples

Patient	Key facts	Score	Red flag	Priority	Primary action
A, 78	DFU left hallux, ABPI 0.62, area +18% in 2 weeks, open 54 days	20 + 20 + 0 + 15 + 3 + 2 = 60	RF-01	SEE TODAY	Same-day diabetic foot MDT and vascular referral

Patient	Key facts	Score	Red flag	Priority	Primary action
B, 84, care home	Sacral pressure ulcer, newly recorded Category 4	25 + 8 + 0 + 0 + 0 + 4 = 37	RF-03	SEE TODAY	TVN review today
C, 69	VLU, delayed healing (4-week PAR 15%), ABPI 0.95, open 140 days	10 + 12 + 0 + 0 + 6 + 0 = 28	None	SEE THIS WEEK (floor)	TVN review this week
D, 72	Surgical wound, suspected infection, stable size, open 20 days	12 + 5 + 9 + 0 + 0 + 1 = 27	None	SEE THIS WEEK (floor)	Review for infection this week
E, 81	VLU, 4-week PAR 55%, no infection, open 45 days	10 + 0 + 0 + 0 + 3 + 0 = 13	None	HEALING	Continue plan; consider reduced visits
F, 66	Category 2 heel ulcer, stable, open 12 days	8 + 5 + 0 + 0 + 0 + 1 = 14	None	MONITOR	Continue plan

Patient A shows why red flags come first: a score of 60 alone would give SEE THIS WEEK, but DFU with PAD is limb-threatening and goes to the top of today's list.

9.10 Requirements

ID	Requirement	Priority
PRI-01	Every patient shall have exactly one priority from the four bands, or 'Unable to prioritise'	Must
PRI-02	Band thresholds shall be absolute and shall not vary with caseload size or distribution	Must
PRI-03	Any fired red flag shall set SEE TODAY, overriding score	Must
PRI-04	Each prioritisation shall record engine version, ruleset version and input hash	Must
PRI-05	Re-running the engine on the same inputs and versions shall produce identical outputs	Must
PRI-06	Ruleset changes shall require Clinical Safety Officer approval within the application	Must
PRI-07	Prioritisation of a 5,000-patient tenant shall complete in under 2 minutes after consolidation	Should

ID	Requirement	Priority
PRI-08	Tenants shall be able to run a proposed ruleset in shadow mode and compare outcomes before activation	Should

10. Clinical confidence engine

Clinical Confidence tells the clinician how far to trust a priority, rated HIGH, MEDIUM or LOW from a 0 to 100 score built on assessment count, data completeness, trajectory reliability and recency. Confidence never lowers a priority; low confidence adds a data-capture action and a visible warning.

10.1 Components

Component	Max	How it is scored
Assessment count	25	Assessments with usable area on the driving wound in the last 12 weeks: 1 → 5; 2 → 12; 3 → 18; 4 → 22; 5 or more → 25
Data completeness	30	Critical fields present: 15 (all or nothing per field, proportionally); important fields present for the wound type: up to 15. For DFU and leg ulcers, a vascular assessment within 12 weeks is an important field
Trajectory reliability	30	Valid (not interpolated) 2- and 4-week windows: 10 each; slope fit with 3 or more points: 5; no implausible values excluded: 5
Recency	15	Latest assessment ⩽ 7 days: 15; 8 to 14 days: 10; 15 to 28 days: 5; > 28 days: 0

```
C = C_{count} + C_{complete} + C_{traj} + C_{recency}
```

10.2 Levels and caps

Level	Score	Display	Behaviour
HIGH	⩾ 75	Solid indicator	Priority shown without caveat
MEDIUM	50 to 74	Half indicator, limiting factor named	Data-capture action added if a critical or important field is missing
LOW	< 50	Hollow indicator, 'Limited data' label	Data-capture action always added; patient flagged for review if priority is MONITOR or HEALING

Caps apply after scoring:

- **Single assessment:** maximum LOW.
- **Any critical field missing** (wound type, assessment date, location): maximum MEDIUM.
- **Latest assessment older than 28 days:** maximum MEDIUM, and an 'overdue reassessment' supporting factor.
- **Identity unconfirmed:** maximum LOW.

10.3 Why low confidence never lowers priority

A wound with sparse data is not a safe wound; it is an unknown one. If confidence reduced priority, the patients with the poorest documentation, often the most vulnerable, would sink down the list. Instead, LOW confidence on a MONITOR or HEALING patient raises a 'Review: limited data' marker, and on SEE TODAY or SEE THIS WEEK it simply tells the clinician what to check first.

10.4 Example

Patient C (Section 9.9): 4 assessments (22), completeness 24 (ABPI present, tissue type missing on 2 entries), trajectory 25 (valid 2- and 4-week windows, slope fitted, no exclusions), recency 10 (11 days). Score 81, HIGH confidence.

Patient B (Section 9.9): 1 assessment (5), completeness 26, trajectory 0, recency 15. Score 46, capped LOW. The priority stays SEE TODAY because the red flag does not depend on history; the explanation adds 'First recorded assessment: trajectory not yet available'.

11. Explainability and AI clinical reasoning

Every patient row and detail view shows the same seven elements, generated from the Prioritisation Event, so a clinician can see why a priority was given in under 10 seconds and challenge it with specifics. No priority is shown without its explanation.

Element	Definition	Generation rule
Current Status	Trajectory class of the driving wound, plus wound count	Trajectory engine
Overall Risk	The priority band and score	Prioritisation engine
Primary Clinical Driver	The single fact that most determined the priority	Highest red flag; else highest-scoring component
Supporting Factors	Up to 4 further facts, most important first	Remaining score components above 3 points, data gaps, other wounds
What Changed Since Last Review	Differences since the last clinician-confirmed review	Diff of input snapshots; new measurements, state changes, new red flags
Recommended Action	One primary action with timeframe	Action library (9.8)
Clinical Confidence	HIGH, MEDIUM or LOW with limiting factor	Confidence engine

11.1 Language rules

1. Clinical vocabulary a TVN would use: 'area reduced 22% in 4 weeks', never 'risk score delta −0.22'.
2. Numbers with units and timeframes; dates as day and month.
3. Source attribution for each fact (assessment date and recorder role), one tap away.
4. No speculative wording such as 'likely to amputate'; state facts and the rule they triggered.
5. Data drawn from free text is labelled 'from notes'.

11.2 Example: SEE TODAY

```
MRS A, 78  ·  NHS 943 476 5919  ·  Team North              SEE TODAY
-----------------------------------------------------------------------
Current status   Deteriorating (1 open wound: DFU, left hallux)
Overall risk     SEE TODAY  ·  red flag RF-01  ·  score 60
Primary driver   Diabetic foot ulcer with PAD (ABPI 0.62 on 14 Oct)
Supporting       Area up 18% in 2 weeks (3.9 to 4.6 cm2)
                 Slough up from 30% to 50%
                 Open 54 days; no MDT referral recorded
What changed     ABPI recorded for the first time (0.62)
                 Area increase since review on 30 Sep
Recommended      Same-day referral to diabetic foot MDT and vascular
Confidence       HIGH (5 assessments; toe pressure not recorded)
```

11.3 Example: SEE THIS WEEK

```
MR C, 69  ·  Team South                                SEE THIS WEEK
-----------------------------------------------------------------------
Current status   Delayed healing (venous leg ulcer, right gaiter)
Overall risk     SEE THIS WEEK  ·  score 28  ·  floor: delayed healing
Primary driver   Area reduced 15% in 4 weeks (expected 40% or more)
Supporting       Open 140 days
                 No TVN review in 74 days
                 ABPI 0.95 (12 Aug): compression appropriate
What changed     4-week reduction slowed from 21% to 15%
Recommended      TVN review this week; review treatment plan
Confidence       HIGH (4 assessments, 11 days since last)
```

11.4 Example: HEALING

```
MRS E, 81  ·  Team North                                  HEALING
-----------------------------------------------------------------------
Current status   Improving (venous leg ulcer, left gaiter)
Overall risk     HEALING  ·  score 13
Primary driver   Area reduced 55% in 4 weeks (expected 40%)
Supporting       No signs of infection; exudate reduced to low
What changed     Area 5.1 to 2.3 cm2 since 16 Sep
Recommended      Continue plan; consider reducing visit frequency
Confidence       MEDIUM (3 assessments; last measured 16 days ago)
```

11.5 Requirements and acceptance criteria

ID	Requirement	Priority
EXP-01	Every displayed priority shall show all seven elements	Must
EXP-02	Every supporting factor shall link to its source assessment	Must
EXP-03	'What changed' shall compare against the last clinician-confirmed review, not the last engine run	Must
EXP-04	The explanation shall be reproducible from the stored Prioritisation Event	Must
EXP-05	Explanations shall be exportable as plain text for pasting into an EPR note or referral	Should

- **Given** a SEE TODAY patient, **when** a nurse opens the row, **then** the primary driver names the red flag in plain clinical terms and the source assessment date.
- **Given** a patient whose priority moved from MONITOR to SEE THIS WEEK, **when** viewed, **then** 'What changed' states the specific new fact that caused the move.

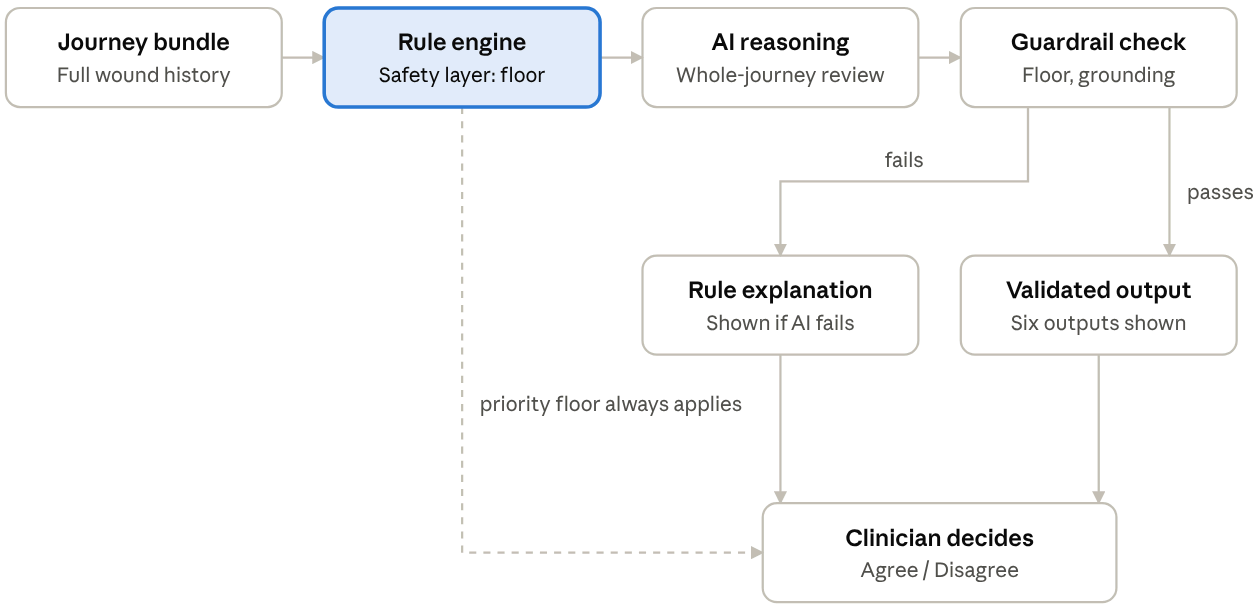
11.6 AI Clinical Reasoning Engine: hybrid reasoning

WoundCareIQ uses hybrid reasoning: the hard-coded clinical rules (Sections 8 to 10) remain the safety layer and set the minimum priority, and an AI Clinical Reasoning Engine sits on top to read the patient's whole wound journey the way an experienced TVN would. The AI explains, contextualises and can escalate; it can never lower a priority set by the rules or override a red flag.

Layer	What it does	Can it change the priority?	Failure behaviour
Rule engine (safety layer)	Red flags, score, bands, floors, confidence	Sets the priority floor	Deterministic; 'Unable to prioritise' if inputs fail

Layer	What it does	Can it change the priority?	Failure behaviour
AI Clinical Reasoning Engine	Reviews the full history; writes status, driver, change, action and clinical narrative	May raise by one band, with cited evidence; never lowers	Falls back to the rule-based explanation (11.2 to 11.4)
Guardrail validator	Checks every AI output against the data and the rules	Rejects outputs that break a rule	Rejected output is logged and replaced by the rule-based explanation
Clinician	Reviews and decides	Always	Agree / Disagree feedback (Section 16)

Rules set the floor; AI explains and can only escalate



Hybrid prioritisation flow · rules, AI, guardrails

The rule priority travels straight to the clinician whatever the AI does; the AI's output reaches the screen only after passing every guardrail, otherwise the rule-based explanation is shown.

11.6.1 Design principles

1. **Rules first, always.** The AI runs only after the rule engine has produced a priority. Its input includes that priority as a fixed fact.
2. **The floor is absolute.** Final priority = the higher of the rule priority and any validated AI escalation. If the AI judges a patient less urgent than the rules, it may say so in the narrative as 'for TVN consideration', and the priority does not move.
3. **Whole journey, not a score.** The AI reasons over the full assessment history and previous priorities, not the clinical score. It is asked how the wound has behaved, what has been tried and how it responded.
4. **Every claim is grounded.** Every fact in the output must cite an assessment, risk factor or prioritisation event ID from the evidence bundle. Uncited or unverifiable claims cause rejection.
5. **Clinical language, clinical limits.** The output reads like a TVN case review. It does not diagnose beyond the data, choose dressings or drugs, or predict outcomes such as amputation.

11.6.2 Inputs: the wound journey bundle

The engine does not receive raw EPR records. A deterministic builder assembles a pseudonymised, structured bundle per patient, so the AI sees exactly the facts the rules saw, in time order.

Input	Content in the bundle
Assessment history	Every assessment for every open wound in the last 12 months, dated, with source ID
Wound measurements	Length, width, depth, area, tissue percentages, exudate, undermining, with excluded values marked
Wound trajectory	Class per window (2 to 12 weeks), PAR values, slope, benchmark for the wound type
Infection history	Infection state timeline, antimicrobial courses, swab results, escalations
PAD	ABPI, TBPI, toe pressures, pulses and Doppler findings with dates; PAD tier

Input	Content in the bundle
Diabetes	Diabetes type, foot service referral status, DFU history
Pressure ulcer category	Category over time, location, equipment and repositioning plan where recorded
Chronicity	Onset date (recorded or estimated), duration band, recurrence history
Previous priorities	Every prioritisation event with band, drivers, acknowledgements and TVN feedback verdicts
Rule output (fixed)	Current rule priority, red flags fired, score components, confidence and limiting factors
Care context	Care setting, care goal (healing, maintenance, palliative), interventions started (for example compression, offloading, debridement) with dates

Names, NHS numbers, addresses and free-text identifiers are removed before the bundle leaves the clinical store; the AI sees an internal patient key, age band and sex only.

11.6.3 Reasoning framework: how the AI reviews a patient

The engine is instructed to follow the same sequence an experienced TVN uses when reviewing notes before a visit:

1. **Aetiology check.** What is this wound, and does the documented history support that? (For example a 'venous' ulcer with ABPI 0.6 is not a simple venous ulcer.)
2. **Journey phases.** Divide the history into phases: onset, response to first plan, plateau or deterioration, response to any change of plan.
3. **Inflection points.** Identify when the trajectory changed and what happened around that date: new infection, compression stopped, equipment change, care-setting move.
4. **Response to interventions.** Did the wound respond as expected to what was done? A wound not improving 4 weeks after compression started is a different problem from one never compressed.
5. **Risk combination.** How do diabetes, PAD, infection and pressure interact for this patient now?
6. **Priority history.** Has this patient been flagged before? Was action taken? Repeated SEE THIS WEEK flags with no change in plan are themselves a reason for escalation.

7. **Why now.** What, specifically, makes today different from the last review?
8. **Next action.** The single most useful next step, consistent with the action library and the rule floor.

11.6.4 Outputs

The engine returns a structured JSON object with six outputs, each validated before display.

#	Output	Content	Constraint
1	Current Status	Trajectory phase in clinical words, for example 'Deteriorating after an 8-week plateau'	Must be consistent with the trajectory engine's class
2	Overall Risk	Final priority band, rule priority, and any AI escalation with its reason	Never below rule priority
3	Primary Clinical Driver	The single fact that most determines urgency	If a red flag fired, must name it
4	What Changed Since Last Review	Specific new facts since the last clinician-confirmed review, with dates and values	Every item cited; none invented
5	Recommended Action	One primary action with timeframe, drawn from the action library, plus up to two secondary actions	Must be at least as urgent as the rule action
6	Clinical Narrative	80 to 150 words answering why this patient, why now, what changed and what next	All numbers and dates must match the bundle

```
{
  "current_status": {"text": "Deteriorating after an 8-week plateau",
"evidence": ["asm_11", "asm_14"]},
  "overall_risk": {"final_priority": "SEE_TODAY", "rule_priority":
"SEE_TODAY", "ai_escalation": null},
  "primary_driver": {"text": "Diabetic foot ulcer with PAD (ABPI 0.62, 14
Oct)", "red_flag": "RF-01", "evidence": ["asm_14"]},
  "what_changed": [
    {"text": "First ABPI recorded: 0.62", "evidence": ["asm_14"]},
    {"text": "Area 3.9 to 4.6 cm2 since 30 Sep", "evidence": ["asm_12",
"asm_14"]}
  ],
  "recommended_action": {"action_code": "URGENT_REFERRAL_DIABETIC_FOOT_MDT",
"timeframe": "today", "secondary": ["RECORD_TOE_PRESSURE"]},
  "clinical_narrative": {"text": "...", "evidence": ["asm_08", "asm_11",
"asm_14", "pe_07"]},
  "why_now": "New ABPI confirms PAD in a DFU that has stopped healing",
  "model_version": "reasoner-1.0.0",
  "prompt_version": "tvn-review-2026.10"
}
```


11.6.5 Example clinical narratives

Mrs A, 78, SEE TODAY (rule priority SEE TODAY, RF-01)

> Mrs A's left hallux DFU improved steadily for its first 5 weeks with offloading, falling from 5.2 to 3.9 cm². Since 30 September it has stopped responding: area is up 18% in 2 weeks and slough has risen from 30% to 50%. The ABPI recorded on 14 October is 0.62, so this is now a diabetic foot ulcer with arterial disease, which explains the loss of progress. No foot service referral has been recorded. She has been SEE THIS WEEK twice since September without a change in plan. She needs same-day referral to the diabetic foot MDT and vascular team, and a toe pressure if it can be obtained, because ABPI can underestimate disease in diabetes.

Mr C, 69, SEE THIS WEEK (rule priority SEE THIS WEEK; no AI escalation)

> Mr C's right gaiter venous leg ulcer has been open 140 days. Compression started on 12 August after an ABPI of 0.95, and the ulcer reduced 21% in the first 4 weeks, then only 15% in the most recent 4 weeks, well below the 40% expected. There are no signs of infection. He has not had a TVN review in 74 days. This is a slowing, hard-to-heal ulcer on appropriate compression rather than an acute problem. A TVN review this week should confirm the aetiology and whether compression is being tolerated and worn as planned.

Example of a validated AI escalation

> Mrs G's sacral Category 3 pressure ulcer is scored SEE THIS WEEK by the rules. Across 6 weeks it has deteriorated three times, each after a period of improvement, and each deterioration followed a recorded mattress change or hospital stay. Previous priorities were SEE THIS WEEK on 3 occasions, each acknowledged, with no change to the pressure-relieving plan recorded. The pattern suggests the plan is not holding. Escalated to SEE TODAY: TVN review today to check equipment and the repositioning plan before the next deterioration.

An AI escalation is shown with a distinct 'AI-raised' marker and its reason, so the clinician can see that the rules alone gave SEE THIS WEEK.

11.6.6 Guardrails and validation of every output

Check	Rule	On failure
Priority floor	Final priority ⩾ rule priority	Reject; use rule explanation
Red-flag consistency	Fired red flags appear in the primary driver or narrative	Reject
Escalation limit	At most one band above rule priority; must cite ⩾ 2 evidence items	Escalation removed; text kept if otherwise valid
Grounding	Every number, date, measurement and category matches the cited evidence	Reject
Citation coverage	Every factual sentence has at least one evidence ID	Reject

Check	Rule	On failure
Action library	Action codes exist in the library and timeframe is not later than the rule action	Reject
Scope	No dressing, drug or dose named; no diagnosis absent from the data; no outcome prediction	Reject
Language	80 to 150 words; NHS clinical terminology; no reassurance wording on SEE TODAY patients	Regenerate once, then reject

If an output is rejected twice, the deterministic explanation (Sections 11.2 to 11.4) is shown with the note 'AI review unavailable'. The patient's priority is unaffected, because the AI never holds it.

11.6.7 Model and hosting

- Large language model accessed through a UK-hosted endpoint with zero data retention and contractual exclusion of training on WoundCareIQ data.
- Pseudonymised bundles only (11.6.2); no direct identifiers leave the clinical store.
- Locked model and prompt versions recorded on every output; deterministic settings (temperature 0); a version change is a ruleset-equivalent change requiring CSO approval, shadow comparison and a hazard-log update.
- Outputs cached per input hash; regenerated only when the bundle changes.
- Latency target: narrative available within 60 seconds of a prioritisation run; the worklist never waits for the AI.

11.6.8 Evaluation of AI reasoning

Measure	Method	Pilot target
Critical factual errors	Every displayed narrative checked by the validator; TVN reports via reason code AI_FACT_ERROR	0 displayed
Minor factual or wording issues	Weekly sample of 50 narratives reviewed by 2 TVNs	< 2%

Measure	Method	Pilot target
TVN trust rating	Blind rating 1 to 5 on accuracy, reasoning, usefulness, trust (AI narrative vs rule template)	Median ⩾ 4; AI preferred over template
Escalation precision	Share of AI escalations the TVN agrees with	⩾ 70%
Missed-context rate	Disagree verdicts where history was relevant and the narrative missed it	Falling month on month
Red-team suite	200 scripted adversarial and edge cases (conflicting data, missing ABPI, palliative goals, debridement) run on every model or prompt change	100% guardrail pass

The AI layer runs in shadow during pilot weeks 5 to 12: narratives are generated and rated but shown only to the evaluation TVNs, and AI escalations are recorded but not applied. It goes live in supported use (weeks 13 to 26) only if the targets above are met.

11.6.9 Requirements

ID	Requirement	Priority
AI-01	The AI engine shall run only after the rule engine and shall receive the rule priority as a fixed input	Must
AI-02	Final priority shall never be lower than the rule priority	Must
AI-03	The AI may raise priority by at most one band, with at least two cited evidence items and an 'AI-raised' marker	Must
AI-04	Every output shall contain the six outputs in 11.6.4, each fact citing evidence IDs	Must
AI-05	Outputs failing any guardrail shall not be displayed; the rule-based explanation shall be shown instead	Must
AI-06	No direct patient identifiers shall be sent to the model	Must

ID	Requirement	Priority
AI-07	Model and prompt versions shall be locked, recorded per output and changed only through clinical governance	Must
AI-08	Clinicians shall be able to flag an AI narrative as factually wrong, triggering CSO review	Must
AI-09	The worklist shall remain fully usable if the AI service is unavailable	Must

- **Given** the rules give SEE TODAY with RF-01, **when** the AI narrative judges the patient stable, **then** the priority stays SEE TODAY and the output is rejected if it omits RF-01.
- **Given** an AI narrative states an area value that differs from the cited assessment, **when** validated, **then** it is rejected and the rule-based explanation is shown.
- **Given** the AI service times out, **when** a nurse opens the worklist, **then** priorities and rule-based explanations display normally with 'AI review unavailable'.

12. TVN workflow

Today a TVN finds high-risk wounds through referrals and memory; with WoundCareIQ the whole footprint is triaged before the first phone call, and referrals arrive with trajectory attached. The time figures below are design hypotheses to be measured in the pilot (Section 18), not claims.

12.1 Current NHS workflow (typical community provider)

1. Referrals arrive by task, email, phone and paper forms, with variable detail.
2. TVN opens each patient in SystmOne or EMIS and scrolls through assessments to rebuild the history.
3. Measurements are compared by hand; photographs, if any, are opened separately.
4. TVN phones the referring nurse for missing information (ABPI, infection signs, current dressing).
5. TVN decides: visit, advise remotely, or redirect (vascular, diabetic foot MDT, GP).
6. Patients never referred stay invisible unless spotted at a team meeting or audit.
7. Reporting for the Director of Nursing is assembled manually from incident data and audits.

Pain points: reactive; triage effort grows with referral volume, not risk; no visibility of unreferred deterioration; inconsistent escalation across teams.

12.2 Future workflow with WoundCareIQ

1. Overnight, assessments are imported, consolidated and prioritised.
2. 08:30: TVN opens the footprint worklist, already sorted SEE TODAY first, each row explained.
3. For each SEE TODAY patient, TVN reads the explanation, opens detail if needed, marks Agree or Disagree, and acts (visit, call the team, refer). The explanation text is copied into the EPR note or referral.
4. SEE THIS WEEK patients are allocated to the week's case reviews and joint visits.
5. Low-confidence patients generate data-capture tasks for the caseload holder (for example 'record ABPI').
6. District nurse team leads use the team view at the daily allocation huddle.
7. Monthly, the Director of Nursing receives an automated report: time to review for SEE TODAY patients, agreement rate, healing rates by wound type.

12.3 Comparison

Step	Current	With WoundCareIQ	Hypothesised saving
Finding high-risk patients	Wait for referral	Whole caseload screened nightly	Unreferred high-risk patients found, measured weekly
Rebuilding history per patient	10 to 15 min of EPR review	Trajectory and drivers on one screen	5 to 10 min per patient
Chasing missing data	Phone calls after triage	Data gaps flagged to caseload holder before triage	Fewer call-backs
Daily allocation huddle	Discussion from memory and lists	Agenda from team worklist	Shorter, more consistent huddles
Board reporting	Manual audit	Automated monthly	1 to 2 days of audit per month

12.4 Clinical impact (to be tested)

- Earlier specialist input for DFU with PAD and spreading infection, measured as time from first qualifying assessment to MDT or vascular referral.
- Fewer wounds passing 12 weeks without specialist review.
- More consistent escalation across teams, measured as variation in referral rates for comparable risk.
- Safe step-down of visit frequency for HEALING patients, freeing capacity, monitored for any re-deterioration within 4 weeks.

13. API design

All functionality is exposed through a versioned REST API (`/v1`) over TLS 1.2+, authenticated with OAuth 2.0 / OIDC (NHS CIS2 for clinical users; client credentials for system integrations), and scoped per tenant. The web application uses the same API as integrators.

13.1 Conventions

- JSON request and response bodies; ISO 8601 timestamps in UTC; NHS numbers as 10-digit strings.
- Every response carries `X-Request-Id` ; every write is recorded in the audit log.
- Pagination by cursor (`?cursor=&limit=` , max 200).
- Errors follow RFC 9457 problem details.
- Idempotency keys (`Idempotency-Key` header) on all POSTs.
- Rate limits: 600 requests per minute per user; 60 per minute per integration for imports.

13.2 Endpoint summary

Area	Method and path	Purpose	Roles
CSV import	`POST /v1/imports`	Upload a CSV with a mapping profile; returns an import job	Data admin, integration
CSV import	`GET /v1/imports/{importId}`	Job status, row counts, validation errors	Data admin

Area	Method and path	Purpose	Roles
CSV import	`GET /v1/imports/{importId}/errors`	Row-level rejected and flagged records	Data admin
Mapping	`GET /v1/mapping-profiles`	List versioned mapping profiles	Data admin
Patients	`GET /v1/patients`	Worklist; filter by priority, facility, wound type, confidence	Clinical roles
Patients	`GET /v1/patients/{patientId}`	Patient detail with wounds, trajectory, current explanation	Clinical roles
Patients	`GET /v1/patients/{patientId}/history`	Prioritisation events over time	Clinical roles
Wounds	`GET /v1/wounds/{woundId}/assessments`	Assessment history for a wound	Clinical roles
Wounds	`POST /v1/wounds/{woundId}/merge` · `/split`	Correct wound grouping	TVN
Prioritisation	`POST /v1/prioritisation/runs`	Trigger a run for a tenant or facility	System, data admin
Prioritisation	`GET /v1/prioritisation/runs/{runId}`	Run status and summary	Data admin
Prioritisation	`POST /v1/prioritisation/evaluate`	Evaluate one patient payload without storing (testing, shadow rulesets)	Integration, clinical safety
Prioritisation	`POST /v1/patients/{patientId}/acknowledgements`	Record 'Plan in place' on a red flag	TVN
Reporting	`GET /v1/reports/caseload-summary`	Counts by priority and facility	Manager, DoN
Reporting	`GET /v1/reports/outcomes`	Healing rates, time to review, escalation metrics	DoN, ICB (aggregate)
Reporting	`GET /v1/reports/validation`	Agreement rates by rule, band, user group	Clinical safety, product
Audit	`GET /v1/audit-events`	Query audit log by user, patient, action, date	Auditor, IG lead

Area	Method and path	Purpose	Roles
Feedback	`POST /v1/patients/{patientId}/feedback`	Agree or Disagree with the current priority	Clinical roles
Feedback	`GET /v1/feedback`	Review feedback queue	Clinical safety, product

13.3 CSV import

```
POST /v1/imports
Content-Type: multipart/form-data
Idempotency-Key: 6f1d2c8e-...

file=@assessments_2026-10-14.csv
source=systmone
mapping_profile=humber-s1-v3
mode=incremental
```

```
{
  "import_id": "imp_01HG2",
  "status": "processing",
  "received_rows": 4812,
  "links": {"self": "/v1/imports/imp_01HG2"}
}
```

Completed status returns `accepted_rows`, `rejected_rows`, `flagged_rows`, `duplicates_dropped`, `new_patients`, `identity_queue` and `prioritisation_run_id`. Rejection reasons are coded (for example `E101_INVALID_NHS_NUMBER`, `E204_UNMAPPED_WOUND_TYPE`, `W310_IMPLAUSIBLE_AREA`).

13.4 Patient retrieval

```
GET /v1/patients?
priority=SEE_TODAY,SEE_THIS_WEEK&facility=fac_team_north&limit=50
```

```
{
  "data": [
    {
      "patient_id": "pat_01HF8Z3K9Q",
      "display_name": "Mrs A",
```

```
      "age": 78,
      "priority": "SEE_TODAY",
      "score": 60,
      "red_flags": ["RF-01"],
      "current_status": "DETERIORATING",
      "primary_driver": "Diabetic foot ulcer with PAD (ABPI 0.62)",
      "recommended_action": "Same-day referral to diabetic foot MDT and
vascular",
      "confidence": "HIGH",
      "open_wounds": 1,
      "last_assessed_at": "2026-10-14T10:32:00Z",
      "acknowledged_until": null
    }
  ],
  "next_cursor": "eyJvIjo1MH0"
}
```

13.5 Prioritisation evaluate (stateless)

Used by integrators and the clinical safety team to test rules. Accepts the canonical patient bundle (patient, wounds, assessments, risk factors) and returns the full Prioritisation Event without persisting it. An optional `ruleset_version` allows shadow comparison.

13.6 Feedback

```
POST /v1/patients/pat_01HF8Z3K9Q/feedback
```

```
{
  "prioritisation_event_id": "pe_01HG2A7M",
  "verdict": "DISAGREE",
  "clinician_priority": "SEE_THIS_WEEK",
  "reason_codes": ["ALREADY_UNDER_MDT", "DATA_OUT_OF_DATE"],
  "comment": "Seen by MDT yesterday, plan in place"
}
```

13.7 Audit logs

Every read of identifiable data and every write produces an audit event: `event_id`, `timestamp`, `actor_id`, `actor_role`, `tenant_id`, `action` (for example `PATIENT_VIEW`, `FEEDBACK_SUBMIT`, `RULESET_ACTIVATE`), `resource`, `patient_id` where applicable,

`source_ip`, `request_id`. Audit events are append-only and exported nightly to write-once storage.

13.8 Webhooks and FHIR (Phase 2+)

- `priority.changed` and `redflag.fired` webhooks for integration into task lists.
- A FHIR R4 facade exposing priorities as `Flag` and recommendations as `Task`, with `Observation` references to source assessments.

13.9 AI reasoning

Method and path	Purpose	Roles
`GET /v1/patients/{patientId}/ai-review`	Current validated AI review (six outputs) or the rule-based fallback with reason	Clinical roles
`POST /v1/ai-reviews/{aiReviewId}/flags`	Report an AI factual error or unsafe wording (reason code AI_FACT_ERROR)	Clinical roles
`POST /v1/ai-reasoning/evaluate`	Run the AI layer on a test bundle with a chosen model and prompt version, without storing	Clinical safety, evaluation
`GET /v1/reports/ai-quality`	Rejection rate by guardrail, escalation agreement, trust ratings	Clinical safety, product

14. Database design

The operational store is PostgreSQL 16 with row-level security per tenant; assessments are append-only, engine outputs are immutable, and every table carries `tenant_id`. Identifiable fields (NHS number, name, date of birth, postcode) are encrypted at column level with tenant-specific keys; analytics use a separate pseudonymised warehouse.

14.1 Entity relationships

```
tenant 1──* facility 1──* patient 1──* wound 1──* assessment
                              |           |
                              |           └──* clinical_event (wound-level)
                              ├──* risk_factor
                              ├──* prioritisation_event 1──1 recommendation
                              |           |               1──1
clinical_confidence
                              |           └──* feedback
                              ├──* acknowledgement
                              └──* patient_source_ref
ruleset 1──* prioritisation_event        user *──* facility (role
assignments)
audit_event (append-only, references any of the above)
```

14.2 Core tables

```
CREATE TABLE facility (
  facility_id      text PRIMARY KEY,
  tenant_id        text NOT NULL,
  parent_id        text REFERENCES facility,
  type             text NOT NULL CHECK (type IN
('locality','community_team','care_home','clinic')),
  name             text NOT NULL,
  ods_code         text
);

CREATE TABLE patient (
  patient_id           text PRIMARY KEY,
  tenant_id            text NOT NULL,
  nhs_number_enc       bytea,
  nhs_number_hash      text,             -- keyed hash for matching
  nhs_number_verified  boolean NOT NULL DEFAULT false,
  name_enc             bytea,
  dob_enc              bytea,
  sex                  text,
  facility_id          text REFERENCES facility,
  care_setting         text,
  identity_status      text NOT NULL DEFAULT 'confirmed',
  current_event_id     text,
  created_at           timestamptz NOT NULL,
```

```
  updated_at            timestamptz NOT NULL,
  UNIQUE (tenant_id, nhs_number_hash)
);

CREATE TABLE wound (
  wound_id                text PRIMARY KEY,
  tenant_id               text NOT NULL,
  patient_id              text NOT NULL REFERENCES patient,
  wound_type              text NOT NULL,
  snomed_code             text,
  location_side           text,
  location_site           text,
  onset_date              date,
  onset_date_source       text,
  status                  text NOT NULL CHECK (status IN
('open','healed','closed_other')),
  pu_category             text,
  recurrence_of_wound_id  text REFERENCES wound,
  care_goal               text DEFAULT 'healing'
);

CREATE TABLE assessment (
  assessment_id     text PRIMARY KEY,
  tenant_id         text NOT NULL,
  wound_id          text NOT NULL REFERENCES wound,
  source_system     text NOT NULL,
  source_record_id  text NOT NULL,
  mapping_profile   text NOT NULL,
  content_hash      text NOT NULL,
  assessed_at       timestamptz NOT NULL,
  entered_at        timestamptz,
  assessor_role     text,
  length_cm numeric, width_cm numeric, depth_cm numeric,
  area_cm2 numeric, area_method text,
  tissue jsonb, exudate jsonb, infection_signs text[],
  infection_status_recorded text,
  pain_score smallint, vascular jsonb,
  quality_flags text[], free_text_enc bytea,
  ingested_at       timestamptz NOT NULL,
  UNIQUE (tenant_id, source_system, source_record_id, content_hash)
);

CREATE TABLE risk_factor (
```

```
  risk_factor_id  text PRIMARY KEY,
  tenant_id       text NOT NULL,
  patient_id      text NOT NULL REFERENCES patient,
  type            text NOT NULL,
  severity        text,
  evidence        jsonb NOT NULL,
  valid_from      date NOT NULL,
  valid_to        date,
  derived         boolean NOT NULL
);

CREATE TABLE clinical_event (
  clinical_event_id  text PRIMARY KEY,
  tenant_id          text NOT NULL,
  patient_id         text NOT NULL REFERENCES patient,
  wound_id           text REFERENCES wound,
  event_type         text NOT NULL,  -- INFECTION_ESCALATED, REDFLAG_FIRED,
REFERRAL_RECORDED, HEALED, RECURRED
  occurred_at        timestamptz NOT NULL,
  detail             jsonb
);

CREATE TABLE prioritisation_event (
  event_id            text PRIMARY KEY,
  tenant_id           text NOT NULL,
  patient_id          text NOT NULL REFERENCES patient,
  computed_at         timestamptz NOT NULL,
  engine_version      text NOT NULL,
  ruleset_version     text NOT NULL REFERENCES ruleset,
  input_snapshot_hash text NOT NULL,
  priority            text NOT NULL,
  previous_priority   text,
  clinical_score      smallint,
  red_flags           text[],
  explanation         jsonb NOT NULL,   -- seven elements, Section 11
  driving_wound_id    text REFERENCES wound
);

CREATE TABLE recommendation (
  recommendation_id text PRIMARY KEY,
  event_id          text NOT NULL UNIQUE REFERENCES prioritisation_event,
  action_code       text NOT NULL,
  action_text       text NOT NULL,
```

```
  secondary_actions text[],
  due_by            timestamptz
);

CREATE TABLE clinical_confidence (
  confidence_id     text PRIMARY KEY,
  event_id          text NOT NULL UNIQUE REFERENCES prioritisation_event,
  level             text NOT NULL CHECK (level IN ('HIGH','MEDIUM','LOW')),
  score             smallint NOT NULL,
  components        jsonb NOT NULL,
  limiting_factors  text[]
);

CREATE TABLE feedback (
  feedback_id         text PRIMARY KEY,
  tenant_id           text NOT NULL,
  event_id            text NOT NULL REFERENCES prioritisation_event,
  user_id             text NOT NULL,
  user_role           text NOT NULL,
  verdict             text NOT NULL CHECK (verdict IN ('AGREE','DISAGREE')),
  clinician_priority  text,
  reason_codes        text[],
  comment_enc         bytea,
  created_at          timestamptz NOT NULL
);
```

Supporting tables: `tenant`, `user_account`, `role_assignment`, `patient_source_ref`, `acknowledgement`, `ruleset` (versioned JSON rules, approver, activation dates), `mapping_profile`, `import_job`, `import_row_error` and `audit_event` (append-only, partitioned by month).

14.3 Design decisions

Decision	Rationale
Append-only assessments; corrections as new rows with `supersedes` link	Clinical record integrity and reproducible trajectories
Immutable prioritisation events	Any historical worklist can be reconstructed for incident review
Keyed hash of NHS number for matching	Matching without decrypting identifiers

Decision	Rationale
JSONB for tissue, exudate, vascular	Template variation across sites without schema churn
Row-level security on `tenant_id`	Hard tenant isolation enforced in the database, not only the application
Retention: per the NHS Records Management Code of Practice as applied by each controller; WoundCareIQ deletes or returns data at contract end	Processor obligations under UK GDPR

15. Product requirements and user stories

The v1 product has six user-facing modules: worklist, patient detail, feedback, data operations, reporting and administration. Requirements below add to the engine requirements in Sections 7, 9 and 11; 'Must' items are required for the Phase 1 pilot.

15.1 Functional requirements

ID	Module	Requirement	Priority
FR-01	Worklist	Show one row per patient, grouped by priority band in the order SEE TODAY, SEE THIS WEEK, Unable to prioritise, MONITOR, HEALING	Must
FR-02	Worklist	Each row shows name, age, facility, priority, primary driver, recommended action, confidence and last assessment date	Must
FR-03	Worklist	Filter by facility, team, wound type, priority, confidence, red flag and acknowledgement status	Must
FR-04	Worklist	Band counts shown as numbers of patients, never as percentages	Must
FR-05	Worklist	Rows that changed band since the user's last visit are marked 'new' or 'moved up'	Should
FR-06	Patient detail	Show all seven explanation elements, every open wound with its trajectory, and assessment history	Must

ID	Module	Requirement	Priority
FR-07	Patient detail	Area-over-time chart per wound, with healing benchmark line	Must
FR-08	Patient detail	Copy explanation as plain text for the EPR or a referral	Should
FR-09	Feedback	Agree / Disagree controls on every patient detail and worklist row	Must
FR-10	Feedback	Disagree requires the clinician's priority and at least one reason code	Must
FR-11	Data ops	Upload CSV, choose mapping profile, see validation report before commit	Must
FR-12	Data ops	Identity-confirmation and wound-grouping queues	Must
FR-13	Reporting	Caseload summary, outcomes and validation reports, exportable as CSV and PDF	Must
FR-14	Admin	Manage users, roles, facility assignments and mapping profiles	Must
FR-15	Admin	Ruleset management with draft, shadow, approved and active states and electronic CSO sign-off	Must
FR-16	Notifications	Daily email or in-app digest of new SEE TODAY patients per user scope (no patient identifiers in email)	Could

15.2 Non-functional requirements

ID	Category	Requirement
NFR-01	Availability	99.5% monthly during pilot; 99.9% from Phase 2, excluding agreed maintenance windows
NFR-02	Performance	Worklist of 500 patients loads in under 2 seconds at p95
NFR-03	Freshness	Priorities recomputed within 15 minutes of an import completing

ID	Category	Requirement
NFR-04	Scalability	50 tenants, 250,000 active patients, 10 million assessments without redesign
NFR-05	Recovery	RPO 1 hour, RTO 4 hours; tested twice a year
NFR-06	Accessibility	WCAG 2.2 AA; NHS design system patterns; colour never the only carrier of priority
NFR-07	Devices	Current Chrome, Edge and Safari; usable on NHS-issued tablets at 768 px width
NFR-08	Hosting	UK data residency for all patient data, backups and logs
NFR-09	Security	Annual CREST-accredited penetration test; critical findings fixed within 14 days
NFR-10	Determinism	Same inputs and versions give identical outputs
NFR-11	Observability	Engine run metrics, import error rates and API latency monitored with alerting

15.3 Epics and user stories

Epic 1: Daily triage

- As a TVN, I want to open a footprint worklist sorted by clinical severity, so that I see the highest-risk patients first.
  - Acceptance: SEE TODAY patients appear first; within the band, red flags sort above score-only entries; no patient appears twice.
- As a district nurse, I want to filter to my team, so that the daily huddle covers only our caseload.
  - Acceptance: filter persists for the session; band counts update to the filtered set.

Epic 2: Understanding a priority

- As a community nurse, I want to see what changed since the last review, so that I know what to look for at the visit.
  - Acceptance: changes listed with dates and values; empty state reads 'No change since last review on [date]'.

- As a TVN, I want to see each wound's area over time against the expected healing rate, so that I can judge trajectory at a glance.
  - Acceptance: chart shows measured points, excluded points marked, and the benchmark line from the first assessment.

Epic 3: Validation

- As a TVN, I want to mark Agree or Disagree in one tap, so that giving feedback does not slow triage.
  - Acceptance: Agree is one action; Disagree opens a short form with clinician priority and reason codes; both are audited.
- As the Clinical Safety Officer, I want to see agreement rates by rule, so that I can find rules that need review.
  - Acceptance: validation report shows agreement by band, red flag and wound type with counts.

Epic 4: Data operations

- As a data administrator, I want to see row-level validation errors before committing an import, so that bad data does not reach clinicians.
  - Acceptance: import preview lists rejected and flagged rows with codes; commit is blocked if critical-field error rate exceeds 5%.

Epic 5: Assurance and reporting

- As a Director of Nursing, I want a monthly report of time from SEE TODAY flag to recorded review, so that I can assure the board.
  - Acceptance: report shows median and 90th percentile by team, with counts.
- As an ICB analyst, I want anonymised healing-rate metrics by provider, so that I can compare services fairly.
  - Acceptance: small numbers under 10 are suppressed; case-mix shown alongside rates.

16. TVN validation and feedback loop

Every priority can be marked ✅ Agree With Priority or ❌ Disagree With Priority; those verdicts are the pilot's validation dataset, the live safety signal, and the only route by which rules change. The algorithm is locked in production: it never learns automatically from feedback.

16.1 Capturing feedback

Verdict	What the clinician does	What is stored
✅ Agree	One tap	Verdict, event ID, user, role, time
❌ Disagree	Choose their priority; pick one or more reason codes; optional comment	Verdict, clinician priority, reason codes, comment, plus whether the disagreement is over- or under-triage

Reason codes

Code	Meaning	Category
DATA_OUT_OF_DATE	Newer information not yet in the EPR	Data
DATA_INCORRECT	Measurement or field wrong	Data
WRONG_WOUND_GROUPING	Wounds merged or split incorrectly	Data
ALREADY_UNDER_SPECIALIST	Under MDT, vascular or TVN plan	Context
PALLIATIVE_OR_MAINTENANCE	Healing is not the goal	Context
RULE_TOO_SENSITIVE	Clinically not this urgent	Rule
RULE_MISSED_RISK	More urgent than shown	Rule (safety)
OTHER	Free text required	Other

16.2 Feedback loop

1. **Immediate.** A Disagree with under-triage (clinician priority higher than system) creates a safety review item for the Clinical Safety Officer within 1 working day. The clinician's judgement always stands for that patient.
2. **Weekly.** The product and clinical team triage all disagreements: data issues go to data operations (mapping fixes); context issues become candidate data fields (for example care goal); rule issues become candidate rule changes.
3. **Monthly.** A discordance panel (pilot-site TVN lead, WoundCareIQ CSO, clinical informatician) reviews candidate rule changes with evidence.

4. **Release.** An approved change is replayed against all historical prioritisation events, run in shadow mode for at least 2 weeks, signed off by the CSO, recorded in the hazard log, and released as a new ruleset version.

16.3 Trust scoring

Trust is measured per rule, per band and per site, never used to grade individual nurses.

```
\text{Trust}_{rule} = \text{lower bound of the 95\% Wilson interval of agreement for that rule}
```

Using the lower bound stops a rule looking trustworthy on a handful of verdicts.

Trust level	Condition	Action
Trusted	Lower bound ⩾ 0.85, n ⩾ 30	No action
Watch	0.70 to 0.85, or n < 30	Included in monthly panel review
Review	< 0.70 with n ⩾ 30, or any confirmed missed-risk case	Rule reviewed at next panel; CSO may suspend the rule's band effect

The site-level trust dashboard also shows over-triage rate, under-triage rate, the share of verdicts by reason category, and the proportion of priorities that received any verdict (feedback coverage).

16.4 Guarding against bias in the feedback

- **Automation bias:** in the first 4 weeks of the pilot a sample of patients is shown to TVNs without the system priority; the TVN records their own priority first, giving an unanchored comparison.
- **Fatigue:** feedback is requested on all SEE TODAY patients and a rotating 20% sample of other bands, rather than every row every day.
- **Inter-rater variation:** a monthly set of 30 cases is rated independently by two TVNs to measure clinician-to-clinician agreement (Cohen's kappa), the benchmark the system is compared against.

17. Compliance, clinical safety and security

WoundCareIQ must be safe, lawful and assured before any pilot goes live: DCB0129 clinical safety case, DSPT 'Standards Met', DTAC submission, DPIA with each controller, and a documented regulatory position on medical device status. The table summarises obligations; detail follows.

Area	Obligation	Owner	Evidence	Needed by
UK GDPR and DPA 2018	Processor contract, DPIA support, lawful processing	DPO	Data Processing Agreement, DPIA, ROPA	Pilot start
NHS DSPT	Annual supplier self-assessment	IG lead	'Standards Met' publication	Pilot start
DCB0129	Manufacturer clinical risk management	Clinical Safety Officer	Clinical Risk Management Plan, hazard log, Clinical Safety Case Report	Pilot start; every release
DCB0160	Deploying organisation's clinical risk management	Pilot site CSO, supported by WoundCareIQ	Site hazard log, deployment safety case	Pilot start
DTAC	NHS digital technology assessment	Head of Product	Completed DTAC	Pilot procurement
Medical device regulation	Qualification and classification under UK MDR 2002	Regulatory lead	Intended-purpose statement, classification rationale, QMS plan	Before Phase 2 commercial use

Area	Obligation	Owner	Evidence	Needed by
Cyber security	Cyber Essentials Plus; ISO 27001 roadmap; penetration testing	CTO	Certificates, test reports	Pilot start (CE+), Phase 2 (ISO 27001)
Accessibility	WCAG 2.2 AA	Design lead	Accessibility statement and audit	Pilot start

17.1 UK GDPR

- **Roles.** Each NHS provider is the data controller; WoundCareIQ is processor for direct care. Any use for product improvement or analytics beyond the controller's purposes requires separate agreement and is done on anonymised or pseudonymised data.
- **Lawful basis (controller).** Typically Article 6(1)(e) public task and Article 9(2)(h) health or social care; confirmed in each controller's DPIA.
- **National data opt-out.** Not applicable to direct care; applied to any secondary use such as ICB analytics on identifiable or pseudonymised data, unless the data is anonymised.
- **Data minimisation.** Only fields in Section 5.1 are ingested; free text is encrypted and access-restricted.
- **Data subject rights.** Supported through the controller; processor assists within 5 working days.
- **International transfers.** None. All data, backups and logs remain in the UK.
- **Caldicott principles.** Role-based access ensures each user sees only patients they have a legitimate relationship with.

17.2 NHS Data Security and Protection Toolkit

WoundCareIQ completes the DSPT annually as a supplier, achieving 'Standards Met' before processing any patient data. Supporting controls include staff data security training, a named SIRO and Caldicott Guardian advisor, incident reporting within 24 hours to the controller, and business continuity plans tested yearly.

17.3 Clinical safety: DCB0129 and DCB0160

DCB0129 (manufacturers) and DCB0160 (deploying organisations) are mandatory information standards for health IT that could affect patient care, and explicitly cover clinical decision support. NHS England ran a consultation on revising both standards from 29 June to 11 September 2026, prompted by AI and connected systems (NHS England consultation). WoundCareIQ will comply with the current versions (DCB0129 Amd 24/2018, DCB0160 Amd 25/2018) and track the revision.

WoundCareIQ clinical safety management system

1. A named Clinical Safety Officer: a registered clinician (TVN) with clinical safety training.
2. Clinical Risk Management Plan covering the full product lifecycle.
3. Hazard log, reviewed at every ruleset or engine change and monthly during the pilot.
4. Clinical Safety Case Report issued per release, shared with each deploying site.
5. Safety incident process linked to the feedback loop (Section 16).

Initial hazard log extract

ID	Hazard	Cause	Possible harm	Controls
H-01	High-risk patient shown as low priority	Missing PAD data; rule gap; mapping error	Delayed treatment, amputation	Red-flag override; low confidence never lowers priority; data-gap actions; under-triage safety review
H-02	Patient missing from worklist	Import failure; identity mismatch	Patient not reviewed	Import reconciliation counts; 'Unable to prioritise' band; identity queue
H-03	Two patients merged	Wrong NHS number in source	Wrong clinical decisions	No merge without validated NHS number; split function; audit
H-04	Over-reliance on system	Automation bias	Clinician defers to wrong priority	Decision-support wording; unanchored sampling; training; EPR remains source of truth

ID	Hazard	Cause	Possible harm	Controls
H-05	Stale data shown as current	Delayed export	Decisions on old information	Data freshness banner; last-assessment date on every row
H-06	Alert fatigue	Too many SEE TODAY patients	Real red flags ignored	Clinically tuned thresholds; acknowledgement with expiry; monitoring of band sizes
H-07	Wrong ruleset activated	Configuration error	Systematic mis-prioritisation	CSO electronic sign-off; shadow mode; versioned rollback

Additional hazards for the AI Clinical Reasoning Engine

ID	Hazard	Cause	Possible harm	Controls
H-08	Narrative states a false fact	Model hallucination	Wrong clinical decision	Grounding and citation checks; rejection with rule fallback; clinician error flag
H-09	Narrative reassures about a high-risk patient	Model weighs history over current red flag	Delayed action	Priority floor; red-flag consistency check; no reassurance wording on SEE TODAY
H-10	Excess AI escalations	Over-sensitive reasoning	Alert fatigue	One-band limit; two-evidence rule; escalation precision monitored

ID	Hazard	Cause	Possible harm	Controls
H-11	Silent behaviour change	Model or prompt updated	Systematic shift in narratives or escalations	Locked versions; CSO approval; red-team suite; shadow comparison
H-12	Identifiable data exposed to model provider	Pseudonymisation failure	Privacy breach	Identifier stripping with automated tests; UK zero-retention endpoint; DPIA

17.4 Medical device status

MHRA guidance treats decision-support software that applies automated reasoning to individual patient data as usually being a medical device (DAC Beachcroft summary of MHRA guidance). WoundCareIQ ranks individual patients by clinical risk, so it should be planned as Software as a Medical Device under UK MDR 2002, likely above Class I, pending formal regulatory advice. MHRA has also published draft Medical Devices (Amendment) Regulations 2026 for comment, which may change software classification (Artixio).

Plan: confirm the intended-purpose statement and classification with a regulatory consultant before the pilot (OQ-2); run Phase 1 as a governed clinical evaluation in shadow mode under the provider's oversight; implement an ISO 13485-aligned QMS and IEC 62304 software lifecycle from Phase 1; achieve UKCA marking before commercial deployment in Phase 2. Device status does not replace DCB0129 and DCB0160.

The AI Clinical Reasoning Engine is part of the device's intended purpose and must be covered by the same classification, clinical evaluation and post-market surveillance, including any AI-specific MHRA requirements in force at the time. Keeping the rule engine as the deterministic safety layer, and limiting the AI to explanation and bounded escalation, is designed to keep the AI's clinical risk controllable and auditable.

17.5 Cyber security

- Cyber Essentials Plus before pilot; ISO 27001 certification during Phase 2; alignment with NCSC cloud security principles.

- Encryption: TLS 1.2+ in transit; AES-256 at rest; column-level encryption for identifiers with keys in a managed HSM-backed key service.
- Network: private subnets, WAF, no public database endpoints, least-privilege service identities.
- Secure development: code review, dependency and container scanning, SAST in CI, secrets scanning, signed builds.
- Annual CREST penetration test plus a test before each major release; vulnerability SLAs (critical 14 days, high 30 days).
- Logging and monitoring with 24/7 alerting; incident response playbook; controller notified within 24 hours of a personal-data incident.

17.6 Audit trails

- Every view of identifiable data, every feedback verdict, acknowledgement, wound merge or split, ruleset change and export is logged (Section 13.7).
- Logs are append-only, retained for at least 8 years or as the controller specifies, and exported to write-once storage.
- Each prioritisation can be reproduced from its input hash and versions for incident investigation.
- Monthly access audits by the provider's IG lead are supported by a built-in report of unusual access (out-of-scope patients, bulk viewing).

17.7 Role-based access

Role	Patient data scope	Can do	Cannot do
Community / district nurse	Own team(s)	View worklist and detail; give feedback	Change rules; view other teams
Care home nurse	Own facility	View facility worklist; give feedback	View community caseloads
TVN	Assigned footprint	All clinical views; acknowledge red flags; merge or split wounds	Activate rulesets

Role	Patient data scope	Can do	Cannot do
Wound specialist / surgeon	Referred patients	View summary and detail; feedback	Browse caseloads
Director of Nursing	Organisation (aggregate; identifiable on drill-down, audited)	Reports; drill-down	Edit clinical data
Provider manager	Organisation (pseudonymised)	Operational reports	View identifiable records
ICB analyst	Anonymised aggregates only	Population reports	Any patient-level data
Data administrator	Import metadata, error rows	Imports, mapping profiles, identity queue	Clinical feedback
Clinical Safety Officer	As TVN plus validation data	Approve rulesets; manage hazard log	—
Auditor / IG lead	Audit log	Query audit events	Clinical views

Authentication uses NHS CIS2 smartcard or authenticator where available, otherwise organisational SSO with enforced MFA. Sessions time out after 15 minutes of inactivity.

18. NHS pilot design

The Phase 1 pilot runs for 6 months at one NHS community provider, across 3 community nursing teams, the TVN service and 2 nursing homes, using CSV exports and starting in shadow mode. It has to prove one thing first: that WoundCareIQ does not miss patients a TVN would see today, and second, that it saves time and is trusted.

18.1 Objectives

1. **Safety:** demonstrate high sensitivity for SEE TODAY patients against TVN judgement.
2. **Validity:** show agreement with TVNs at least as good as TVN-to-TVN agreement.
3. **Operational value:** reduce time spent on triage and surface unreferred high-risk patients.

4. **Adoption:** reach routine daily use by TVNs and team leads.
5. **Readiness:** produce the evidence pack (safety case, validation report, DTAC, case study) for Phase 2 and ICB-wide rollout.

18.2 Stakeholders

Stakeholder	Role in pilot
Provider Director of Nursing	Executive sponsor; chairs steering group
TVN lead	Clinical lead; discordance panel member
District nurse team leads (3)	Daily users; huddle integration
Care home managers (2)	Facility users
Provider CSO and IG lead / DPO	DCB0160, DPIA, data sharing
Provider informatics / BI team	Scheduled SystmOne and EMIS exports
ICB nursing and digital leads	Observers; scale-up decision
Health Innovation Network	Independent evaluation support and spread
Academic partner	Evaluation design, statistical analysis
Patient and public representatives	Review of communications and evaluation questions
WoundCareIQ product, CSO, informatician, engineering	Delivery, training, support

18.3 Phases and gates

1. **Set-up (weeks −6 to 0).** DPIA, DSPT, DCB0129 and DCB0160 signed off; mapping profile validated (98% agreement on critical fields); retrospective replay on 12 months of historical data. *Gate: CSO sign-off at both organisations.*
2. **Unanchored baseline (weeks 1 to 4).** TVNs prioritise a daily sample of patients without seeing WoundCareIQ output; the system runs silently. *Gate: SEE TODAY sensitivity ⩾ 95% on baseline cases.*
3. **Shadow mode (weeks 5 to 12).** Priorities shown to TVNs alongside usual practice; every SEE TODAY and a 20% sample of others receive Agree or Disagree. No change to visit allocation driven by the tool. *Gate: overall agreement ⩾ 80%, no unresolved under-*

triage safety issue.

4. **Supported use (weeks 13 to 26).** Worklist used at daily huddles and TVN triage, with clinicians still deciding; outcome and operational metrics collected. *Gate: steering-group decision on Phase 2.*

18.4 Success metrics

Domain	Metric	Target
Safety	Sensitivity of SEE TODAY against TVN reference	⩾ 95%
Safety	Confirmed missed-risk incidents attributable to the tool	0
Validity	Agreement rate on reviewed priorities	⩾ 85% by week 26
Validity	Weighted kappa, system vs TVN, compared with TVN vs TVN	Not inferior
Clinical outcome	Median time from first qualifying assessment to MDT or vascular referral for DFU with PAD	Reduced vs 12-month baseline
Clinical outcome	Wounds open > 12 weeks with no TVN review in 60 days	Reduced by 30%
Clinical outcome	Proportion of leg ulcers with ABPI within 12 weeks of onset	Increased vs baseline
Clinical outcome	Re-deterioration within 4 weeks after HEALING step-down	⩽ baseline rate
Operational	TVN minutes per day on caseload triage (time-and-motion)	Reduced by 30%
Operational	Unreferred high-risk patients identified per month	Reported (no target)
Operational	Import success rate; data freshness	⩾ 98%; < 24 hours
Adoption	TVN weekly active use	100% of pilot TVNs
Adoption	Team leads using worklist at daily huddle	⩾ 4 days a week per team

Domain	Metric	Target
Adoption	Feedback coverage of SEE TODAY patients	⩾ 90%
Experience	User satisfaction (SUS score)	⩾ 70

Longer-term outcomes such as amputation rates and wound-related admissions are recorded but the pilot is not powered to show change in them; they are the focus of the Phase 2 multi-site evaluation.

18.5 Validation strategy

1. **Retrospective replay.** Run the engine on 12 months of historical data; review a stratified sample of 300 patients with a two-TVN panel; check whether known adverse outcomes (amputation, admission with wound infection, Category 4 deterioration) were preceded by SEE TODAY or SEE THIS WEEK flags.
2. **Prospective unanchored comparison.** Weeks 1 to 4 baseline gives sensitivity and specificity per band without anchoring bias.
3. **Prospective agreement.** Shadow-mode Agree / Disagree data gives agreement by rule and wound type, analysed with trust scoring (Section 16.3).
4. **Inter-rater benchmark.** Monthly dual-rated case sets establish TVN-to-TVN kappa.
5. **Independent evaluation.** The Health Innovation Network and academic partner analyse outcome and operational data and publish a pilot report.
6. **Governance route.** The provider and academic partner confirm with the HRA decision tool whether the evaluation is service evaluation or research; if research, HRA approval is obtained before week 1.

19. Roadmap

The roadmap moves from a manual CSV pilot to direct EPR feeds, then to prediction and population analytics; each phase starts only when the previous phase's gate is met, so safety evidence always leads functionality.

Each phase opens only after its gate evidence is met



WoundCareIQ roadmap · 5 phases, 4 gates

The Phase 1 gate uses the pilot targets in Section 18.4; later gates depend on supplier access (OQ-1) and device conformity (OQ-2).

Phase	Indicative duration	Key deliverables	Users and sites	Exit evidence
1 · CSV upload, manual pilot	6 months pilot plus 2 months set-up	Import and mapping, consolidation, trajectory, prioritisation, confidence, explainability, worklist, feedback, reports	1 provider, 3 teams, 2 care homes	Validation report, safety case, DTAC, independent evaluation
2 · SystmOne integration	9 to 12 months	Supplier-approved feed, near-real-time updates, webhooks, FHIR facade, multi-tenant onboarding tooling	3 to 5 providers in one or two ICBs	Interface live, UKCA marking, ISO 27001
3 · EMIS integration	6 to 9 months	EMIS Web feed; cross-setting patient view; care-home system connectors	Practices and providers in pilot ICBs	Linked outcome dataset
4 · Predictive deterioration	12 months	Model predicting deterioration 2 to 4 weeks ahead, shown as an advisory signal next to rule-based priority; model card; drift monitoring	Validated sites only	Prospective validation; regulatory update

Phase	Indicative duration	Key deliverables	Users and sites	Exit evidence
5 · Population health analytics	Ongoing	ICB outcome dashboards, provider benchmarking, equity analysis, NWCSP alignment reporting	ICBs, Health Innovation Networks	Commissioning adoption

Principles for Phase 4. The predictive model never replaces red-flag rules or the four bands; it adds a 'likely to deteriorate' signal with its own explanation (the factors behind the prediction) and confidence. It is trained only on data the controllers have agreed to, versioned and locked, and changed only through the same governance as rules (Section 16.2).

AI Clinical Reasoning Engine timing. The AI layer (Section 11.6) is built in Phase 1 and runs in shadow during pilot weeks 5 to 12, going live in supported use only if its evaluation targets are met. Phase 4 prediction is separate: it forecasts future deterioration, whereas the reasoning engine explains the patient's current position.

20. Investor summary

Wound care is one of the NHS's largest and least-optimised clinical costs, almost all of it in the community, and no product today turns the data already recorded into a daily decision about who to see. WoundCareIQ is that layer: low-friction to deploy because it reads existing EPR data, defensible because its value grows with validated clinical rules and longitudinal outcome data.

20.1 Market opportunity

- **Scale of spend:** £8.3 billion a year on wound management in the UK NHS, 81% in the community; £5.6 billion on wounds that did not heal in the year (Guest et al, 2020).
- **Workforce pressure:** 54.4 million district and community nurse visits a year are wound-related; any tool that safely redirects even a small share of visits frees substantial capacity.

- **Buyers:** NHS community providers (primary), ICBs (population analytics), care-home groups, and later acute trusts and international community health systems with similar EPR-centric data.
- **Policy pull:** the National Wound Care Strategy Programme recommendations, ICB accountability for population outcomes, and NHS emphasis on shifting care into the community all favour tools that make community care measurable.

20.2 Clinical problem in one line

Community teams have the data to spot deteriorating wounds, but not the time or tools to read it, so high-risk patients surface late and specialist time is spent on wounds that are already healing.

20.3 Competitive landscape and differentiation

Category	Examples	What they do	WoundCareIQ difference
EPRs	SystmOne, EMIS Web	Record assessments and care plans	Interprets their data longitudinally and ranks patients; complements rather than competes
Specialist wound EHRs	Net Health Wound Care	Structured wound documentation, analytics within their platform	Source-agnostic layer across all systems in a footprint
Image-capture apps	Smartphone wound measurement and photography tools	Accurate measurement and images at the bedside	Uses their measurements as inputs; focuses on prioritisation and explanation
BI dashboards	Local Power BI builds	Activity counts	Clinical rules, trajectory, confidence and validation, maintained under DCB0129

Defensibility

1. **Clinical ruleset and validation evidence.** TVN-authored, versioned, validated across sites; each pilot strengthens it.
2. **Mapping library.** Every site mapping for SystmOne and EMIS templates lowers the cost of the next deployment.
3. **Longitudinal outcome dataset.** Prioritisation events linked to outcomes are the basis for the Phase 4 predictive model, which competitors without deployed footprint cannot train.
4. **Assurance moat.** DCB0129, DSPT, DTAC and medical device conformity take time and expertise to achieve.
5. **Trust by design.** Explainability and the Agree / Disagree loop drive clinical adoption, where black-box tools stall.

20.4 Business model (working assumptions for validation)

- Annual subscription per community provider, priced per open-wound caseload band, with an implementation fee per source system.
- ICB population analytics module as an add-on licence.
- Value case per provider built from avoided unhealed-wound cost, nurse visits released and avoided escalations; to be quantified from pilot data, not assumed.

20.5 Scalability

- **Technical:** multi-tenant, UK-hosted, mapping-profile driven; onboarding a new site on an already-mapped template targets under 4 weeks.
- **Clinical:** one ruleset with governed site configuration, not bespoke builds.
- **Commercial:** land with one provider, expand to the ICB footprint, then replicate across ICBs using the same EPR mappings.
- **International:** the engine is terminology-based (SNOMED CT); markets with structured community wound data are later expansion options.

20.6 Why WoundCareIQ can become the clinical intelligence layer for wound care

EPRs will keep collecting data; they are built for documentation, not interpretation. Image-capture tools improve measurement but not decisions. The layer in between, which reads everything, applies TVN-grade reasoning, explains itself and is clinically assured, is

unoccupied. WoundCareIQ is designed to occupy it, and every deployment adds mappings, validated rules and outcome data that make the next one easier and the product harder to replicate.

21. Risks, open questions and glossary

The largest risks are data access, medical device classification and clinical trust; each has an owner and a mitigation in place before the pilot.

21.1 Delivery risks

Risk	Likelihood	Impact	Mitigation	Owner
Supplier interface access for SystmOne or EMIS delayed or costly	High	High	CSV route proven in Phase 1; early supplier engagement; FHIR facade ready	CTO
Medical device classification higher than planned	Medium	High	Early regulatory advice; QMS from Phase 1; shadow-mode clinical evaluation	Regulatory lead
Poor source data quality at pilot site	High	Medium	Data-gap actions; confidence engine; mapping validation; data quality reporting to the provider	Clinical informatician
Low clinical trust or adoption	Medium	High	TVN co-design; explainability; Agree / Disagree loop; huddle integration	Head of Product
Alert fatigue from too many SEE TODAY patients	Medium	High	Clinically tuned thresholds; acknowledgement; monitoring band sizes	CSO
Pilot site capacity to support exports and evaluation	Medium	Medium	Funded backfill; automated scheduled exports	Pilot lead

Risk	Likelihood	Impact	Mitigation	Owner
Revised DCB0129 / DCB0160 change requirements	Medium	Low	Track consultation outcome; safety case structured for update	CSO

21.2 Open questions

- [ ] OQ-1: Which supplier-approved interface routes and commercial terms apply for SystmOne community and EMIS Web wound data?
- [ ] OQ-2: What is WoundCareIQ's medical device classification under UK MDR 2002 for the stated intended purpose?
- [ ] OQ-3: Which pilot provider, teams and care homes, and which care-home record systems are in use there?
- [ ] OQ-4: Are pilot benchmark values (Section 8.3) and red-flag thresholds (Section 9.2) accepted by the pilot TVN lead as written?
- [ ] OQ-5: Is the evaluation a service evaluation or research under the HRA decision tool?
- [ ] OQ-6: How should palliative and maintenance care goals be captured where the EPR does not record them?

21.3 Glossary

Term	Meaning
ABPI	Ankle Brachial Pressure Index; ratio of ankle to arm systolic pressure, used to detect PAD
TBPI	Toe Brachial Pressure Index; more reliable than ABPI when arteries are calcified
DFU	Diabetic foot ulcer
PAD	Peripheral arterial disease
VLU	Venous leg ulcer
PU	Pressure ulcer, graded Category 1 to 4, unstageable or deep tissue injury
MDT	Multidisciplinary team; here the diabetic foot service

Term	Meaning
PAR	Percentage area reduction
TVN	Tissue Viability Nurse
CSO	Clinical Safety Officer under DCB0129 / DCB0160
DSPT	Data Security and Protection Toolkit
DTAC	Digital Technology Assessment Criteria
ICB	Integrated Care Board
NWCSP	National Wound Care Strategy Programme
MUST	Malnutrition Universal Screening Tool
SaMD	Software as a Medical Device

21.4 Sources

- Guest JF, Fuller GW, Vowden P. Cohort study evaluating the burden of wounds to the UK's NHS in 2017/2018. BMJ Open 2020
- Guest et al 2020, Table 10: cost per healed and unhealed wound
- NHS England: national review of clinical risk management standards DCB0129 and DCB0160 (consultation, 2026)
- NHS Standards Directory: DCB0160 (Amd 25/2018)
- DAC Beachcroft: when is software regulated as a medical device (MHRA guidance summary)
- Artixio: UK AI medical device regulations and 2026 draft amendments

Clinical thresholds in Sections 8 and 9 (healing benchmarks, ABPI and toe-pressure cut-offs, infection continuum, NICE NG19 referral timeframes) reflect widely used UK practice and international guidance; they are draft values for pilot TVN sign-off (OQ-4), not independently verified for this document.